

LEARNING HETEROGENEOUS TABULAR TASKS THROUGH MULTIREGIME PRETRAINING OF PRIOR-DATA FITTED NETWORKS

Anonymous authors

Paper under double-blind review

ABSTRACT

Tabular datasets can combine observations from different populations, environments, or operating conditions, each governed by a different relationship between features and labels. These coexisting relationships define distinct regimes. Regime membership, however, may be unobserved, requiring classifiers to account for heterogeneity within a single dataset. Prior-data fitted networks (PFNs) are pretrained on synthetic datasets to predict new observations from labelled examples without task-specific retraining. We investigate whether incorporating regime heterogeneity into this pretraining enables compact PFNs to learn multiregime classification. We extend TabICL’s structural-causal-model prior with regime-dependent score functions and label mappings, and compare fixed-mixture and curriculum pretraining with canonical NanoTabPFN pretrained on the native single-regime prior, published tabular foundation models, and conventional classifiers. On held-out synthetic tasks without a supplied regime ID, the largest effect occurs in feature-dependent, multiregime multiclass tasks: curriculum pretraining improves a six-layer NanoTabPFN’s excess cross-entropy from -0.1562 to -0.1730 nats, a 10.8% relative reduction. Across 2,880 matched soft-gate episodes, the paired cross-entropy difference is -0.0168 nats (95% episode-bootstrap interval $[-0.0179, -0.0158]$) and the accuracy-gain difference is $+0.84$ points ($[+0.76, +0.91]$). Across all matched multiregime tasks with three to five classes, the curriculum model exceeds majority-class accuracy by 6.69 points, compared with 6.30 for the native single-regime model, a 6.2% relative increase; its excess cross-entropy improves from -0.0838 to -0.0926 nats, a similarly sized 10.5% relative reduction. Fixed-mixture pretraining produces a similar pooled improvement, so these experiments do not isolate an advantage of the curriculum schedule. Evaluation on 36 real-world BeyondArena tasks and a merged multi-site heart-disease case study finds no consistent advantage over the native single-regime control.

1 INTRODUCTION

Tabular datasets can combine observations from different populations, environments, or operating conditions whose feature-label relationships differ. We refer to these coexisting relationships as regimes. Regime membership may be unobserved, and an observed group need not identify a distinct label mechanism. Conditional-mixture models describe this setting through regime-specific prediction rules (Jacobs et al., 1991; Jordan & Jacobs, 1994). Given a labelled support set and the features of query rows, the prediction problem is to use the support examples to account for this heterogeneity when predicting query labels.

Prior-data fitted networks (PFNs) learn this support-to-query operation by pretraining on synthetic tasks (Müller et al., 2022; Hollmann et al., 2023). They amortize inference: after pretraining, one set of weights maps a labelled context and query features to predictive probabilities without fitting new weights for each task. The pretraining prior specifies the feature-label relationships, class balances, and task variation the model encounters. We build on the structural-causal-model (SCM) prior supplied by TabICL (Qu et al., 2025), which generates features and continuous scores that are

transformed into class labels. NanoTabPFN provides a compact predictor for studying changes to this task distribution (Pfefferle et al., 2025).

Drawing multiple label rules within a pretraining task could teach a PFN to use support examples to resolve competing explanations for a query. We ask whether compact PFNs can learn multiregime classification and whether a multiregime prior improves over a matched native single-regime prior. We separately ask whether any improvement transfers to real tabular tasks with independent and identically distributed (IID), grouped, or temporal splits.

We extend the native SCM prior with regime-dependent scores or score-to-class mappings and train two- to six-layer NanoTabPFN variants with fixed-mixture or curriculum pretraining. The predictor retains its ordinary architecture and receives no regime ID in the main experiment. Within each model size, we hold the configuration and training recipe fixed and compare against pretraining on the native single-regime task stream. All models are evaluated on identical synthetic episodes, with effects separated by routing and outcome type; real-data evaluation uses 36 selected BeyondArena tasks (Purucker et al., 2026).

We evaluate this intervention across model sizes, routing families, and rule mechanisms on matched synthetic episodes, and separately test transfer to selected real tabular tasks and a real-data group-label case study; Section 5 reports the results.

Our contributions are: (1) an extension of TabICL’s structural-causal-model prior with regime-dependent scores (r_z) or score-to-class mappings (g_z), under soft-gate (feature-dependent) and persistent (latent) regime assignment; (2) a matched comparison of native single-regime, fixed-mixture, and curriculum multiregime pretraining across three NanoTabPFN sizes on identical synthetic evaluation episodes; (3) a frozen-checkpoint diagnostic isolating what a supplied, aligned regime tag can add to prediction at matched input width; and (4) real-data transfer checks on 36 selected BeyondArena tasks and a merged multi-site heart-disease case study with an observed, non-synthetic group label.

2 RELATED WORKS

TabPFN and its later generations (Hollmann et al., 2023; 2025; Grinsztajn et al., 2026), TabICL and TabICLv2 (Qu et al., 2025; 2026), and the smaller NanoTabPFN (Pfefferle et al., 2025) span different architectures, scales, and pretraining distributions. Comparisons across these systems provide performance reference points but cannot isolate the effect of changing a task prior. Our study focuses on multiregime extensions of TabICL’s native SCM prior within a matched NanoTabPFN configuration. This intervention is distinct from TabICLv2’s broader changes to generation, architecture, and training.

Classical mixtures of experts use gates to allocate prediction among explicit expert models (Jacobs et al., 1991; Jordan & Jacobs, 1994). Their treatment of conditional heterogeneity motivates our task construction, but our mixture is in the data-generating prior: the predictor has no architectural expert gate. Identifiability analyses make clear that recovering mixture components depends on the model and its assumptions (Jiang & Tanner, 1999); Section 3 details how this limits our interpretation of the predictor’s scores.

Analyses of transformer in-context learning connect prediction from context to implicit Bayesian inference (Xie et al., 2022) and study learning in regression mixtures (Pathak et al., 2024; Jin et al., 2025). These works motivate inference over coexisting task relationships, though their settings and guarantees do not directly cover our nonlinear SCMs, class discretization, or routing conditions.

TableShift and BeyondArena study tabular performance across datasets and evaluation partitions (Gardner et al., 2023; Purucker et al., 2026). This distribution-shift perspective is complementary to our focus on heterogeneity within a task. Grouped or temporal splits can change covariate distributions, sample sizes, and query difficulty without instantiating the regime-dependent label rules in our synthetic prior; Section 4 returns to this distinction when describing our real-data evaluation.

3 MULTIREGIME EXTENSIONS OF THE NATIVE TABICL PRIOR

Let $S = \{(x_i, y_i)\}_{i=1}^n$ be a labelled support set and x_* a query feature vector. A task draws a latent regime $z_i \in \{1, \dots, K\}$ for each row and a regime-dependent conditional rule $p(y_i | x_i, z_i)$. The predictor receives S and x_* , but neither a separate regime ID nor the generating score on the primary evaluation bank. Its output is $p_\theta(y_* | x_*, S)$.

The native control draws SCM features and a continuous score, then TabICL’s `Reg2Cls` transform draws cut points and maps score ranges to class IDs (Qu et al., 2025). We use that one-regime path for the canonical control. For r_z , the SCM emits one score column per regime and the active regime chooses its column; each score receives its own sampled cut points under shared cut-point settings. For g_z , all regimes share the SCM score but draw separate score-to-label maps. Thus two rows with the same score may have different labels. The rg_z training mixture alternates the two mechanisms. With $K = 1$, the native code path is used without a regime-dependent change (Appendix A).

Writing $r_z(x)$ for a regime’s continuous score and h_z for its sampled score-to-class transform, the generated label has the schematic form $y = h_z(r_z(x))$. In r_z tasks both r_z and h_z may vary by regime; in g_z tasks the score is shared and the label transform varies. This distinction keeps the location of heterogeneity explicit. It does not make the regimes statistically identifiable: two different regime constructions can yield similar observable (x, y) samples, and a predictor is scored on labels rather than on the generator’s membership assignments. The task prior also samples SCM geometry and label rules, so the model must adapt from the support set rather than memorize one fixed partition of feature space.

Two assignment families test distinct information conditions (Figure 1). In *soft-gate* tasks, observed x_1 and support-rank cut points determine regime membership, so routing information is present in features. In *persistent* tasks, rows receive stable latent group assignments without an input group ID; these assignments are drawn independently of x at generation time rather than derived from a feature rule, so, unlike soft-gate membership, they cannot in principle be recovered from the query’s features alone. The term gate describes the generator, not an architectural expert gate (Jacobs et al., 1991; Jordan & Jacobs, 1994). Fixed mixing exposes the model to a constant share of multiregime batches; curriculum mixing ramps that share upward over training (Bengio et al., 2009). Both use the same native one-regime branch; Section 4 gives their exact schedules.

The information available at prediction time differs between these families. A soft-gate query can in principle be routed using its observed feature and the labelled support examples, although the cut points themselves are not passed as inputs. Under persistent assignment, the query’s latent group is not a supplied feature; the model must average over that uncertainty or exploit whatever indirect evidence is present. We therefore expect any difference between the families to be informative about the usefulness of observed routing cues, while avoiding the stronger claim that the model reconstructs z . The fixed and curriculum streams expose the model to the same two multiregime rule types but to different proportions over training time. They do not equalize total multiregime exposure.

4 EXPERIMENTAL SETUP

We train matched native-control, fixed-mixture, and curriculum NanoTabPFN runs at three configurations (small/medium/large): 2 layers/64-dimensional embeddings/4 heads/256-wide MLP, 4/128/4/512, and 6/192/6/768. Thus size changes more than depth. Fixed mixing draws multiregime tasks in 30% of batches; curriculum mixing ramps their share from 0 to 50% over steps 1,000–5,000. All runs use initialization seed 2402, the same 10,000-step recipe and native one-regime task source; the two mixtures alternately use r_z and g_z task dumps (pre-generated synthetic episode files). We report final checkpoints, rather than selecting a favorable test checkpoint. Appendix B gives the recorded training recipe and its provenance limits.

The native control is essential to the estimand: a published PFN may use a different architecture, optimization path, and task prior. For each size, candidate and control checkpoints are evaluated on the same episode IDs, allowing per-episode subtraction before aggregation. The nine native runs share a nominal seed and recipe, but that is one training realization per condition. Differences

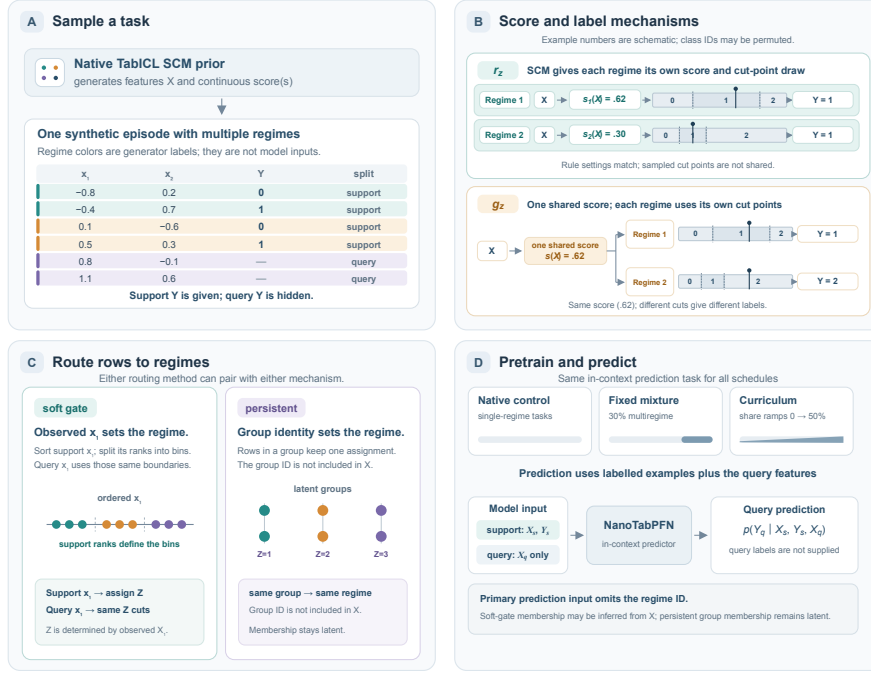


Figure 1: Native task construction. In r_z , regimes use separate SCM scores and independently drawn class cut points; in g_z , they share a score and use separate cut points. Soft-gate membership depends on observed x_1 , while persistent membership is latent. The predictor sees support examples and query features, not regime IDs. The one-regime control, fixed 30% mixture, and 0–50% curriculum share the NanoTabPFN architecture.

across two-, four-, and six-layer configurations describe combined capacity changes because width and MLP size also increase. They should not be read as a depth-only ablation.

The standard-input native test bank varies $K = 1$ –4, binary or 3–5-class labels, binary imbalance, support size, feature count, mechanism, and routing family, including single-rule controls. Matched models see identical episodes. The headline multiregime multiclass subset contains 5,760 episodes in 720 factorial cells; its soft-gate and persistent halves have 2,880 episodes each. We also evaluate 36 selected BeyondArena tasks (27 binary, nine multiclass), with official folds and equal task weighting (Purucker et al., 2026). Appendix F documents those tasks and the independent evaluation protocol.

We define the headline subset before interpreting its results: $K \geq 2$, a multiregime rule, and three to five classes. It contains both r_z and g_z mechanisms and both routing families, with eight episodes per factorial cell. The broader bank additionally contains binary, one-regime, and shared-rule controls; pooling all of them can obscure the behavior on the targeted tasks. BeyondArena evaluation is a separate transfer check on selected real datasets rather than a second sample from the synthetic prior; its evaluation partitions (IID, grouped, temporal) need not correspond to distinct latent label mechanisms, and its split summaries have especially few grouped and temporal tasks, which constrains their interpretation.

The factorial bank enforces a minimum of 32 labelled support rows per requested regime. Consequently, not every regime count occurs at every support size: since each regime needs at least 32 support rows, $K = 4$ occurs only at support sizes 128, 256, and 512 (all but the smallest, 64), while $K = 2$ occurs at every support size. We do not interpret a raw K trend as an isolated effect of regime count. The $K = 1$ branch checks behavior on native one-rule tasks, and shared-rule cells check the geometry of multiple assignments without changing the label rule by regime. The multiregime multiclass subset is the direct test of the training intervention, so it is reported separately from these

controls. Matching episode identities across checkpoints removes evaluation-set composition as an explanation for their paired differences.

For each episode we report query cross-entropy (CE), accuracy gain over majority-class prediction (percentage points), and macro one-versus-rest AUC. Excess CE is model CE minus the CE of the empirical support-class-frequency predictor; smaller is better. Cross-entropy is a proper score, although a CE reduction by itself is not a calibration study (Gneiting & Raftery, 2007). Published PFNs and conventional models use the same synthetic episodes in Table 1. The baseline families include random forests, logistic regression, XGBoost, histogram gradient boosting, LightGBM, and CatBoost (Breiman, 2001; Pedregosa et al., 2011; Chen & Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2018). TabPFN v2.6 is identified by its release card (Prior Labs, 2026); Appendix B lists the configured published-model checkpoint names and the limits of their verification.

For native-model effects, we average within-cell paired candidate-minus-control differences and then give factorial cells equal weight. Negative CE differences and positive accuracy differences favor multiregime pretraining. The percentile intervals in Figure 2 resample evaluation episodes within cells; they quantify finite-bank variation conditional on frozen checkpoints and do not represent uncertainty across training seeds. Absolute Table 1 scores answer a complementary question about model ranking on the same subset. The baseline model families were not trained under the matched native recipe, so their gaps should not be interpreted as prior-only effects.

Additional analyses. We run four further checks, reported in Section 5. First, we audit training-label balance for the native single-regime and pooled multiregime task dumps, in both the binary and multiclass setting, to test whether the multiregime advantage could instead reflect a change in class balance rather than regime structure. Second, we stratify the headline multiclass episodes by empirical support-label balance to check whether the routing effect is confined to a particular imbalance regime. Third, a frozen-checkpoint diagnostic appends a regime-tag column under three conditions – hidden (no tag), shuffled (tag independently permuted within support and query), and true (realized membership, tags independently relabelled per episode) – to measure what a supplied, aligned tag can add to prediction at matched input width (Appendix D). Fourth, for a real-data probe with an observed, non-synthetic group label, we merge three same-schema UCI Heart Disease sites into one 797-row dataset with an observed hospital label (Janosi et al., 1989), evaluated under repeated stratified IID folds and a target-site adaptation protocol that keeps 20 stratified labelled target-site rows in support so the held-out site’s category is genuinely observed rather than imputed (Appendix G).

5 RESULTS

Synthetic: matched headline comparison. Table 1 uses the same 5,760 multiregime multiclass episodes for every model. The native single-regime model already improves over the support-frequency predictor: its excess CE is -0.0838 nats and accuracy gain is 6.30 pp. Fixed and curriculum models reach -0.0914 and -0.0926 excess CE – a 9.1% and 10.5% relative reduction from the native control, respectively; curriculum accuracy gain is 6.69 pp, a 6.2% relative increase over the native control’s 6.30 pp. Paired curriculum-minus-control CE is -0.0088 nats ($[-0.0094, -0.0082]$), and accuracy gain is $+0.38$ pp ($[+0.34, +0.43]$). The fixed-mixture changes are -0.0076 nats and $+0.37$ pp. All published PFNs shown outperform the six-layer native variants on both headline metrics, and all six conventional classifiers trail them on this subset.

For reference, TabPFN v3 reaches 1.1031 CE and 50.59% accuracy, compared with 1.1505 and 48.66% for the curriculum model (Grinsztajn et al., 2026). These are subset scores, distinct from full-bank scores in Appendix C. The fixed and curriculum rows are close: on the headline subset their direct CE values differ by 0.0012 nats and their accuracies by 0.01 percentage points – a much smaller separation than either model’s gap to the native control. For instance, random forest and CatBoost have accuracies within roughly one point of the native control, yet their CE values are higher; accuracy and probability quality need not order methods identically.

Feature-dependent multiclass routing. At six layers, soft-gate multiregime multiclass excess CE improves from -0.1562 for the native control to -0.1730 for curriculum training, a 10.8% relative reduction. Across its 2,880 matched episodes, curriculum minus control CE is -0.0168

Table 1: Matched standard-input synthetic multiregime multiclass comparison: 5,760 episodes, 720 cells, 3–5 classes. Direct query means; lower CE and higher accuracy are better. The three NanoTabPFN rows are six-layer final checkpoints.

Model	Query CE (nats) ↓	Accuracy (%) ↑
Random forest	1.2445	47.57
CatBoost	1.2524	47.33
Logistic regression	1.2832	44.54
XGBoost	1.7099	45.44
Histogram gradient boosting	1.7829	45.20
LightGBM	2.5564	44.92
TabICL v1	1.1243	49.48
TabICL v1.1	1.1389	49.02
TabICL v2	1.1168	50.12
TabPFN v2.2	1.1040	50.55
TabPFN v2.6	1.1077	50.45
TabPFN v3	1.1031	50.59
NanoTabPFN, native single-regime	1.1593	48.28
NanoTabPFN, fixed mixture	1.1517	48.65
NanoTabPFN, curriculum	1.1505	48.66

Matched-control effects across routing and class type

Standard-input blind TEST · multiregime tasks ($K = 2-4$) · 2,880 episodes / 360 cells per panel and size

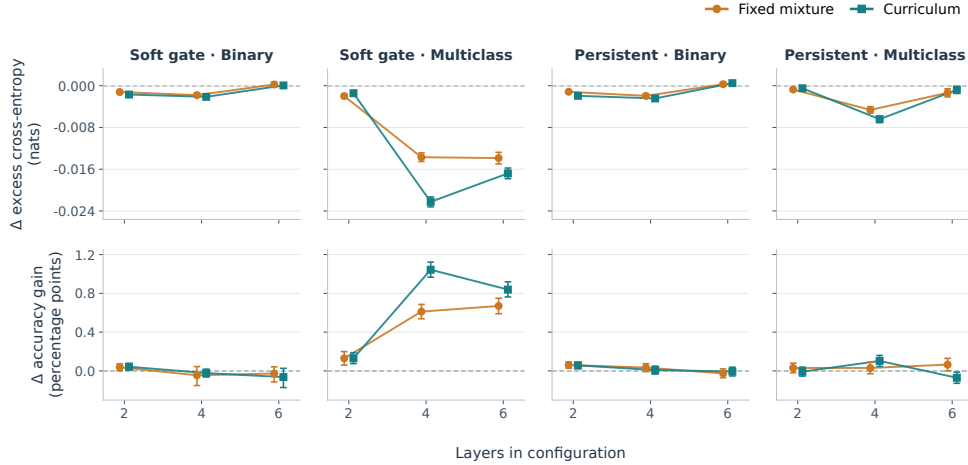


Figure 2: Candidate-minus-same-size-native-control effects on the standard-input test bank, restricted to $K = 2-4$. Each routing-by-class panel includes 2,880 episodes in 360 factorial cells per size. Points equally average cell means; whiskers are 95% percentile intervals from 5,000 within-cell episode-bootstrap replicates. Negative CE and positive accuracy difference favor multiregime pretraining. Intervals omit pretraining-seed variation; model configurations vary in depth, width, and heads.

nats (95% episode-bootstrap interval $[-0.0179, -0.0158]$), and accuracy gain changes by +0.84 pp ($[+0.76, +0.91]$). The corresponding persistent-routing improvement is smaller: fixed mixing changes CE by -0.0014 nats ($[-0.0021, -0.0006]$) and accuracy by +0.07 pp ($[0.00, 0.13]$); curriculum changes them by -0.0008 nats and -0.07 pp.

Rule mechanisms. The improvement is not confined to one way of changing the label rule. Table 2 partitions the six-layer matched multiclass result into r_z and g_z episodes within each routing family. Under soft-gate routing, curriculum changes CE by -0.0164 nats for r_z and -0.0172 for g_z , with accuracy gains of about 0.83–0.85 pp; fixed mixing has smaller but similarly directed effects. In persistent tasks both mechanisms show CE differences below 0.002 nats in magnitude, and

Table 2: Matched six-layer candidate-minus-native effects by routing and rule mechanism on multiregime multiclass episodes. Each row has 1,440 episodes in 180 cells. Negative CE and positive accuracy differences (pp) favor the candidate.

Routing	Rule	Fixed Δ CE	Fixed Δ Acc	Curr. Δ CE	Curr. Δ Acc
soft gate	r_z	-0.0135	+0.67	-0.0164	+0.83
soft gate	g_z	-0.0142	+0.67	-0.0172	+0.85
persistent	r_z	-0.0018	+0.01	-0.0007	-0.07
persistent	g_z	-0.0009	+0.12	-0.0009	-0.07

Table 3: Evaluation-balance sensitivity on the existing standard-input bank. Six-layer candidate minus native-control paired effects; lower CE and higher accuracy differences (pp) favor the candidate. Bins are defined by support majority fraction within each class count, without using outcomes; pooled rows give class counts equal weight. Descriptive fixed-checkpoint contrasts, not a control for training-label balance.

Routing	Maj. bin	n	Cells	Fixed Δ CE	Fixed Δ Acc	Curr. Δ CE	Curr. Δ Acc
soft gate	low	989	340	-0.0154	+0.72	-0.0190	+1.09
soft gate	middle	970	348	-0.0144	+0.65	-0.0175	+0.81
soft gate	high	921	332	-0.0102	+0.42	-0.0132	+0.51
persistent	low	981	335	-0.0029	+0.19	-0.0022	+0.04
persistent	middle	985	348	-0.0020	+0.13	-0.0015	-0.11
persistent	high	914	334	+0.0000	-0.08	+0.0011	-0.12

curriculum accuracy differences are slightly negative. Thus the main routing contrast appears under both sampled rule constructions.

The routing contrast also appears at each requested multiclass count. In soft-gate tasks, curriculum minus native CE is -0.0129 , -0.0162 , and -0.0213 nats for three, four, and five classes, respectively, with 95% episode-bootstrap intervals excluding zero in every case and positive accuracy differences in each slice. In persistent tasks, corresponding CE differences lie between -0.0010 and -0.0004 nats; only the three-class interval excludes zero, and the accuracy direction varies by class count. Each routing-by-class comparison uses 960 matched episodes in 120 cells. The class count changes the scale of CE and the label distribution, so we do not read these magnitudes as a monotone difficulty response.

Limits of the effect. In multiregime binary tasks, the two- and four-layer native variants have positive excess CE, and six-layer models beat the support-frequency baseline by only about 0.003 nats. At six layers, curriculum changes binary CE by $+0.0003$ nats and accuracy by -0.03 pp relative to its control. Four- and six-layer configurations show clearer multiclass changes than the two-layer configurations (Figure 2). Across the entire standard-input test bank, six-layer fixed and curriculum CE is nearly identical (0.7802 and 0.7803).

Evaluation-balance sensitivity. We stratify the same 5,760 episodes by the empirical support majority-class fraction, using tertile cutoffs separately for three, four, and five classes; cutoffs are chosen from labels before looking at model scores, and tied values stay in the lower bin. Table 3 averages paired effects within occupied factorial cells and then gives the three class counts equal weight. The six-layer curriculum soft-gate CE difference remains negative in low, middle, and high support-imbalance bins (-0.0190 , -0.0175 , and -0.0132 nats); fixed mixing is also favorable in each bin. In persistent tasks the effects are much smaller, and the highest-imbalance bin has a curriculum CE difference of $+0.0011$ nats. Appendix C reports all class-count slices and their episode and cell counts.

Training-label balance. In a binary training-dump diagnostic, the mean majority-class fraction is 0.766 for the native single-regime dump and 0.66 for pooled multiregime dumps (Appendix A). The corresponding multiclass comparison across three, four, and five requested classes shows the

same pattern: multiregime dumps are more balanced than the native dump at every class count (Appendix A).

Supplied regime-tag diagnostic. Within each mechanism-by-routing stratum (1,440 matched multiclass episodes, 180 cells), true-minus-shuffled CE ranges from -0.174 to -0.057 nats for persistent r_z and -0.179 to -0.058 for persistent g_z , with accuracy changes of $+4.48$ to $+11.32$ pp and $+4.62$ to $+11.69$ pp. For soft-gate r_z and g_z , CE changes range from -0.062 to -0.034 nats, with accuracy changes of $+1.99$ to $+3.64$ pp and $+2.16$ to $+3.67$ pp. Shuffled-minus-hidden is near zero for persistent tasks (-0.0023 to $+0.0011$ nats) but raises CE by 0.0022 – 0.0057 on soft-gate tasks – an effect of appending a feature, not of its alignment. Appendix D reports the full size-by-model contrasts, their paired intervals, and why these frozen checkpoints cannot support a per-regime recovery analysis.

Real data: BeyondArena transfer. On 36 selected BeyondArena tasks, six-layer curriculum-minus-native-control excess CE is approximately $+0.0036$ nats and accuracy gain -0.10 pp (descriptive differences from rounded task-level values); the fixed mixture is similarly close to the control. Published TabICLv2 and TabPFN v3 outperform the native models on the reported aggregates (Appendix F). The rounded task-level CE comparison is mixed in sign: curriculum is lower on 16 tasks, tied on two at printed precision, and higher on 18. By split, CE differences are $+0.0021$ (32 IID tasks), $+0.0187$ (three grouped), and $+0.0070$ (one temporal).

Merged heart-site case study. In the merged heart-site case study (Appendix G), an observed hospital label has little effect under repeated stratified IID folds: excess CE changes by at most 0.0045 nats across the 20 evaluated models, with no consistent direction. In the target-site adaptation protocol, the aligned-minus-shuffled site-tag contrast in excess CE ranges from -0.0127 to $+0.0042$ across the same 20 models, with 16 of 20 negative; the largest gains are for TabICLv1, random forest, and the medium native configurations. Multiregime pretraining shows no consistent additional advantage over the native single-regime control at any one size.

6 DISCUSSION AND LIMITATIONS

The designed prior produces a measurable but modest synthetic advantage concentrated in feature-dependent multiclass tasks. Soft-gate routing makes regime information available in x_1 ; persistent assignments are not directly supplied. The weak binary results and small persistent-routing effects prevent a broad claim about multiregime classification; as Section 2 notes, this improvement also does not establish reconstruction or identifiability of latent memberships (Limitations). The tag intervention in Appendix D shows that aligned membership can improve prediction, especially for persistent tasks, but it supplies privileged information to the predictor.

The matched synthetic design answers a narrow question: for these checkpoints and this generated distribution, replacing some native one-regime tasks with multiregime tasks helps on selected multiclass evaluations. It does not show that the model uses an explicit routing algorithm or recovers the generator’s row assignments. Near-zero persistent effects indicate little predictive benefit when query membership is unobserved, not that regimes are absent. The supplied-tag diagnostic changes evaluation-time information and tests a different counterfactual.

The training-label balance audit shows multiregime dumps are more balanced in the multiclass setting, so gains could reflect class distribution as well as regime structure; it does not quantify their contributions. Fixed and curriculum mixtures also differ in total multiregime exposure (Section 3), so the comparison supports no specific scheduling explanation. Published PFNs remain stronger on both headline metrics. The soft-gate effect appears in CE and accuracy at six layers, while persistent effects are small and their accuracy direction varies by schedule. This is consistent with useful information in observed x_1 , but predictive scores do not reveal whether the model recovered memberships. Evaluation-balance slices show the soft-gate advantage across imbalance bins, but this descriptive test-bank analysis leaves the training-label confound intact. A decisive comparison would equalize training class balance while varying regime structure and use independent seeds; the current post-hoc slices are exploratory.

Neutral transfer on BeyondArena therefore does not show that real data lack heterogeneity. The same caution applies to accuracy and AUC in that comparison, because different tasks, target types, and split designs contribute to their aggregates. A grouped fold tests prediction for a held-out group; it does not tell us whether the group index is sufficient for the label rule or whether the synthetic prior’s feature-dependent gate is a good approximation. We interpret the real-data results as evidence about these evaluated checkpoints on this selected suite, rather than as validation of the generator’s mechanism.

Equal task weighting prevents larger datasets from dominating, but 36 selected tasks do not represent every tabular setting; Appendix F retains task-level scores. The three grouped tasks split many small groups, averaging as few as 2–12 rows per group, which limits split-specific conclusions. The heart-site intervention shows that an observed group label can affect target-site prediction, but does not recreate an unobserved r_z or g_z process.

The synthetic and real-data evaluations therefore stress different capabilities. In the synthetic soft gate, a designated observed coordinate helps determine membership under the generator. In a real dataset, candidate grouping cues can be distributed among features, absent for new groups, or irrelevant to the conditional label rule. The merged heart data offer one concrete observed-site probe, but the model is scored on disease prediction, not on reconstructing a hidden site or verifying distinct causal mechanisms. Its repeated target-site draws reuse most query rows, so they do not provide 15 independent dataset-level tests. A real-data gain would need to be established on more tasks with documented environment labels and matched training controls. The neutral results here put a bound on what these particular pretraining mixtures achieved, while leaving open whether another prior or architecture would transfer better.

Limitations. All main-model comparisons use one pretraining seed; paired episode-bootstrap intervals quantify test-bank variation for frozen checkpoints, not variation from retraining, model weights, or training streams. Thousands of matched episodes make the fixed-checkpoint CE differences estimable on this bank, but the intervals also do not account for selecting this prior, subset, or diagnostic after looking at earlier experiments. We report CE and accuracy together because a change in predictive probabilities can matter even when the winning class changes rarely; conversely, a small accuracy gain does not show that a model’s probabilities are calibrated. The weak binary and persistent results are part of the same evidence, not exceptions to be discarded in favor of the soft-gate multiclass slice. Predictive improvement anywhere in this study, including under the supplied regime tag, does not establish reconstruction or identifiability of latent memberships. More seeds, broader grouped and temporal coverage, and prior designs closer to real mechanisms are needed before a general transfer claim.

7 CONCLUSION

Multiregime-mixture pretraining improves compact PFN prediction on synthetic multiclass tasks, most clearly when feature values provide routing cues. For the six-layer model on these feature-dependent tasks, curriculum pretraining improves excess cross-entropy by 10.8% relative to the native prior; across all multiregime multiclass tasks, accuracy gain over majority prediction rises from 6.30 to 6.69 percentage points. Fixed-mixture pretraining also improves the pooled benchmark, so the gains are not unique to the curriculum schedule. The selected BeyondArena and merged heart-site evaluations show no consistent transfer advantage. These findings show that PFNs can benefit from priors that include coexisting label rules and motivate further work on multiregime tabular learning.

AI USE STATEMENT

Generative AI assisted with code, experiment execution, analysis, figures, manuscript drafting and editing, tables, and PDF checks, including the balance-slice analysis reported here. The authors report human checks of the manuscript, literature review, tables, experiments, analysis, data preprocessing, code, and figures, and retain responsibility for the final claims and submission. The literature review, data preprocessing, experiment design, and formulation of research ideas were not AI-assisted.

REPRODUCIBILITY STATEMENT

The prior and controls are specified in Sections 3–4 and Appendices A–B; headline synthetic and real-data figures are reported in Section 5, with full task-level tables in Appendices C, F, and G. The September native training used the TabICL SCM from `tabicl` 2.1.1 and TFM-Playground 0.0.1; Appendix B records package and checkpoint identifiers available locally. The balance analysis joins report episode IDs to the saved standard-input test bank without model inference. The 36-task container and official-fold archive, including an immutable BeyondArena revision, is described in Appendix F; the BeyondArena revision hash is provided in the supplementary materials. No anonymous public release is available at drafting time. The saved records omit the historical source commit, checkpoint content hashes, and the 36-task job’s cached dataset revision; the pinned archive matches available task records but does not establish the job’s exact cache identity.

REFERENCES

- Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning*, pp. 41–48. ACM, 2009. doi: 10.1145/1553374.1553380. URL <https://doi.org/10.1145/1553374.1553380>.
- Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324. URL <https://link.springer.com/article/10.1023/A:1010933404324>.
- Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794. ACM, 2016. doi: 10.1145/2939672.2939785. URL <https://arxiv.org/abs/1603.02754>.
- Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7(1):1–30, 2006. URL <https://jmlr.org/papers/v7/demsar06a.html>.
- Josh Gardner, Zoran Popovic, and Ludwig Schmidt. Benchmarking distribution shift in tabular data with TableShift. In *Advances in Neural Information Processing Systems*, volume 36, pp. 53385–53432, 2023. doi: 10.52202/075280-2324. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/a76a757ed479a1e6a5f8134bea492f83-Abstract-Datasets_and_Benchmarks.html.
- Tilman Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437. URL <https://sites.stat.washington.edu/people/raftery/Research/PDF/Gneiting2007jasa.pdf>.
- Léo Grinsztajn, Klemens Flöge, Oscar Key, Felix Birkel, Philipp Jund, Brendan Roof, Mihir Mani, Shi Bin Hoo, Magnus Bühler, Anurag Garg, Dominik Safaric, Jake Robertson, Benjamin Jäger, Simone Alessi, Adrian Hayler, Vladyslav Moroshan, Lennart Purucker, Philipp Singer, Alan Araz, Julien Siems, Jan Hendrik Metzen, Georg Grab, Nick Erickson, Siyuan Guo, Elliott Kalfon, Simon Bing, David Salinas, Clara Cornu, Lilly Charlotte Wehrhahn, Diana Kriuchkova, Kursat Kaya, Lydia Sidhoum, Marie Salmon, Jerry Chen, Madelon Hulsebos, Yann LeCun, Samuel Müller, Bernhard Schölkopf, Sauraj Gambhir, Noah Hollmann, and Frank Hutter. TabPFN-3: Technical report. *arXiv preprint arXiv:2605.13986*, 2026. doi: 10.48550/arXiv.2605.13986. URL <https://arxiv.org/abs/2605.13986>.
- Noah Hollmann, Samuel Müller, Katharina Eggenberger, and Frank Hutter. TabPFN: A transformer that solves small tabular classification problems in a second. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=cp5PvcI6w8_.

- Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirmer, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, 2025. doi: 10.1038/s41586-024-08328-6. URL <https://www.nature.com/articles/s41586-024-08328-6>.
- Robert A. Jacobs, Michael I. Jordan, Steven J. Nowlan, and Geoffrey E. Hinton. Adaptive mixtures of local experts. *Neural Computation*, 3(1):79–87, 1991. doi: 10.1162/neco.1991.3.1.79. URL <https://www.cs.toronto.edu/~hinton/absps/jjnh91.pdf>.
- Andras Janosi, William Steinbrunn, Matthias Pfisterer, and Robert Detrano. Heart disease. UCI Machine Learning Repository, 1989. URL <https://archive.ics.uci.edu/dataset/45/heart+disease>. Dataset.
- W. Jiang and M. A. Tanner. On the identifiability of mixtures-of-experts. *Neural Networks*, 12(9):1253–1258, 1999. doi: 10.1016/S0893-6080(99)00066-0. URL <https://www.sciencedirect.com/science/article/pii/S0893608099000660>.
- Yanhao Jin, Krishnakumar Balasubramanian, and Lifeng Lai. In-context learning for mixture of linear regression: Existence, generalization and training dynamics. *Transactions on Machine Learning Research*, November 2025. URL <https://openreview.net/forum?id=buZXVuTsHY>.
- Michael I. Jordan and Robert A. Jacobs. Hierarchical mixtures of experts and the EM algorithm. *Neural Computation*, 6(2):181–214, 1994. doi: 10.1162/neco.1994.6.2.181. URL <https://doi.org/10.1162/neco.1994.6.2.181>.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL <https://papers.nips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>.
- Samuel Müller, Noah Hollmann, Sebastian Pineda Arango, Josif Grabocka, and Frank Hutter. Transformers can do Bayesian inference. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=KSugKcbNf9>.
- Reese Pathak, Rajat Sen, Weihao Kong, and Abhimanyu Das. Transformers can optimally learn regression mixture models. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=sLkj91HIZU>.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12(85):2825–2830, 2011. URL <https://jmlr.org/papers/v12/pedregosa11a.html>.
- Alexander Pfefferle, Johannes Hog, Lennart Purucker, and Frank Hutter. nanoTabPFN: A lightweight and educational reimplement of TabPFN. *arXiv preprint arXiv:2511.03634*, 2025. doi: 10.48550/arXiv.2511.03634. URL <https://arxiv.org/abs/2511.03634>.
- Prior Labs. TabPFN-2.6 model card. Hugging Face model repository, 2026. URL https://huggingface.co/Prior-Labs/tabpfn_2_6. Official model card; accessed 2026-09-24.
- Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL <https://proceedings.neurips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html>.
- Lennart Purucker, Andrej Tschalzev, Nick Erickson, Gioia Blayer, David Holzmüller, Alan Arazi, Alexander Pfefferle, Mustafa Tajjar, Gaël Varoquaux, and Frank Hutter. Beyond IID: How general are tabular foundation models, really? *arXiv preprint arXiv:2606.30410*, 2026. doi: 10.48550/arXiv.2606.30410. URL <https://arxiv.org/abs/2606.30410>.

Jingang Qu, David Holzmüller, Gaël Varoquaux, and Marine Le Morvan. TabICL: A tabular foundation model for in-context learning on large data. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 50817–50847. PMLR, 2025. URL <https://proceedings.mlr.press/v267/qu25d.html>.

Jingang Qu, David Holzmüller, Gaël Varoquaux, and Marine Le Morvan. TabICLv2: A better, faster, scalable, and open tabular foundation model. In *International Conference on Machine Learning*, 2026. URL <https://arxiv.org/abs/2602.11139>.

Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. An explanation of in-context learning as implicit Bayesian inference. In *International Conference on Learning Representations*, 2022. URL <https://arxiv.org/abs/2111.02080>.

APPENDIX

A NATIVE PRIOR AND IMPLEMENTATION VALIDATION

The native generator draws TabICL SCM hyperparameters using its fixed and sampled defaults, obtains feature columns and one or more continuous score columns, and applies native `Reg2Cls` score-to-class conversion. A sampled score-to-label rule draws class cut points and may permute class IDs. For r_z , each of K score columns receives an independent cut-point draw under the same cut-point settings; for g_z , one score is reused with K independent conversions. The rule active for row i is selected by z_i . With $K = 1$, the implementation follows the native TabICL prior path; the documented validation checked bit-identical output against `tabicl.prior.PriorDataset` for both mechanism settings. The selected labels pass TabICL’s split sanity check, and episodes with non-finite feature values are redrawn. The recorded dump invocation is in `scripts/slurm/dump_multiregime_v4_native.sbatch`; its historical source commit was not logged.

The primary evaluation uses a factorial bank with $K = 1$ –4, binary and 3–5-class outcomes, 0.1/0.3/0.5 controlled binary class-ratio cells, varied support sizes and feature counts, the two rule mechanisms, and soft-gate or persistent assignment. Soft-gate cut points are based on ranks of observed x_1 in the support set. Persistent assignment remains a stable latent group label and is not supplied in the standard input. Multiclass evaluation cells require each requested class to appear. Native pretraining itself produces imbalanced classes. In the documented binary training dumps, the native single-regime tasks have mean majority-class fraction 0.766, with 45% of tasks at or above 0.8; pooling multiregime rules gives 0.66 and 15%, respectively. Per-regime label balance has the native spread, so pooling independent rules moves binary tasks toward balance. Across all 100,000 episodes in each local native single-regime, r_z , and g_z training dump, for three, four, and five requested classes respectively, mean support-majority fractions are 0.638, 0.551, and 0.491 for the native dump; 0.517, 0.432, and 0.376 for r_z ; and 0.515, 0.431, and 0.374 for g_z (12,461/12,444/12,481 episodes per class count, giving episodes equal weight within each requested class count; these describe the source dumps rather than the realized schedule-weighted training stream). Multiregime tasks are therefore more balanced on average in both the binary and multiclass settings; Section 5 (Training-label balance) reports the pooled summary. The earlier non-native generator calibrated classes to fixed balanced vectors and is excluded from this study.

B ARCHITECTURES AND TRAINING REPRODUCIBILITY

Table 4: Recorded NanoTabPFN configurations. These are size configurations, not a depth-only ablation.

Size	Layers	Embedding	Heads	MLP hidden
Small	2	64	4	256
Medium	4	128	4	512
Large	6	192	6	768

Each condition uses seed 2402, 10,000 steps, batch size four with 32 accumulated gradients, learning rate 10^{-4} , minimum 10^{-6} , 2,000-step warmup, weight decay 0.01, and gradient clipping at 1.0. Training tasks have 1,024 rows, 2–12 features, 2–5 classes, and a sampled training fraction between 0.1 and 0.9. The ordinary branch uses the one-regime dump; the multiregime branch alternates r_z and g_z dumps. Fixed mixing uses 30% multiregime batches. The curriculum reaches a 50% share over steps 1,000–5,000. Validation is recorded every 500 steps on the native bank and a separate 16-batch stream; the results here use step-10,000 final checkpoints. The configuration is recorded in `scripts/slurm/pretrain_multiregime_v4_native.sbatch`; the nine saved `config.json` files under `artifacts/cluster_sync/tfm_runs/native/` confirm the size-specific values in Table 4. Their standard-input final-test reports are under `artifacts/cluster_sync/tfm_eval/native_regime_split/blind/`.

The authors confirmed that the project `uv.lock` was the training lock for these September native runs. It records TabICL 2.1.1, TabPFN 8.5.0, and Data Foundry 0.0.5; NanoTabPFN is repository source rather than a separately versioned package. The nine native checkpoint identifiers

are original, rg-z-fixed, and rg-z-curriculum, each at small, medium, and large size, seed 2402, and final step 10,000. The published-model evaluation script names TabICL checkpoint files `tabicl-classifier-v1-20250208.ckpt`, `tabicl-classifier-v1.1-20250506.ckpt`, and `tabicl-classifier-v2-20260212.ckpt`; the Slurm defaults identify TabPFN files `tabpfn-v2.2.ckpt`, `tabpfn-v2.6-classifier-v2.6_default.ckpt`, and `tabpfn-v3-classifier-v3_default.ckpt`. These names identify configured files, not verified content hashes. The synced run records contain configuration and evaluation reports but no native checkpoint binaries or historical source commit. Thus checkpoint content, published file identities, and shared initialization from saved states could not be independently verified from these records.

C ADDITIONAL SYNTHETIC RESULTS

Table 5: Final standard-input test-bank scores pooled across task conditions. Accuracy is a fraction. This full-bank comparison has a different composition from Table 1.

Size	Prior	CE	Excess CE	Accuracy
Small	Native	.8164	−.0192	.6178
	Fixed	.8157	−.0199	.6182
	Curriculum	.8161	−.0195	.6179
Medium	Native	.8095	−.0261	.6304
	Fixed	.8053	−.0303	.6312
	Curriculum	.8036	−.0320	.6317
Large	Native	.7811	−.0545	.6385
	Fixed	.7802	−.0554	.6393
	Curriculum	.7803	−.0553	.6389

Table 5 includes one-regime, shared-rule, binary, and multiclass cells and therefore dilutes the multiregime multiclass headline effect. The standard-input bank is primary. Across multiregime binary cells, the two smaller configurations have positive excess CE, and large configurations have only a small negative excess CE. The large soft-gate multiclass gain and small persistent gain are quantified in Section 5 and Figure 2. Per-cell trajectories on the standard-input and legacy duplicate-X1 banks appear in Figure 3, and training/validation curves for all three sizes appear in Figure 4. The paired analysis uses 5,000 percentile-bootstrap replicates resampled within each factorial cell (seed 20260924); intervals are unadjusted descriptive intervals across evaluation episodes. Fixed mixing and curriculum differ both in temporal schedule and multiregime exposure, preventing a pure scheduling attribution.

Tables 6 and 7 expand the main-text support-balance analysis by requested class count. The support-majority tertile cutoffs are 0.4473 and 0.5313 for three classes, 0.3672 and 0.4375 for four classes, and 0.3164 and 0.3789 for five classes. Discrete support-label counts create ties, so bin episode counts are unequal. Across all nine soft-gate class-by-balance slices the six-layer fixed and curriculum CE differences are negative; persistent differences are smaller and are not uniformly favorable. These are scores for the original bank under its observed label frequencies. No evaluation episodes were regenerated and no model was retrained for this diagnostic.

The companion machine-readable balance report records SHA-256 hashes of the local test bank and verified training lock. Its alignment check compares every report episode’s metadata with the local bank and confirms all binary support and query class fractions. The saved evaluation reports do not retain multiclass labels, so complete label-level equivalence to the unavailable cluster-side HDF5 file cannot be independently checked from these reports alone.

Table 6: Detailed soft gate evaluation-balance slices. Candidate minus same-size native control; accuracy differences are percentage points. Cell means are weighted equally within each class-count slice. Support-majority cutoffs are shared across routing families within class count.

Classes	Routing	Maj. bin	n	Cells	Fixed ΔCE	Fixed ΔAcc	Curr. ΔCE	Curr. ΔAcc
3	soft gate	low	324	113	-0.0112	+0.90	-0.0152	+1.08
4	soft gate	low	348	115	-0.0176	+0.70	-0.0197	+1.08
5	soft gate	low	317	112	-0.0174	+0.56	-0.0221	+1.10
3	soft gate	middle	337	117	-0.0105	+0.39	-0.0139	+0.69
4	soft gate	middle	302	115	-0.0134	+0.64	-0.0154	+0.62
5	soft gate	middle	331	116	-0.0192	+0.93	-0.0233	+1.12
3	soft gate	high	299	109	-0.0051	+0.38	-0.0094	+0.51
4	soft gate	high	310	110	-0.0127	+0.48	-0.0117	+0.44
5	soft gate	high	312	113	-0.0129	+0.39	-0.0186	+0.59

Table 7: Detailed persistent evaluation-balance slices. Candidate minus same-size native control; accuracy differences are percentage points. Cell means are weighted equally within each class-count slice. Support-majority cutoffs are shared across routing families within class count.

Classes	Routing	Maj. bin	n	Cells	Fixed ΔCE	Fixed ΔAcc	Curr. ΔCE	Curr. ΔAcc
3	persistent	low	318	113	-0.0006	+0.09	-0.0021	+0.08
4	persistent	low	331	110	-0.0036	+0.50	-0.0022	+0.21
5	persistent	low	332	112	-0.0046	-0.01	-0.0023	-0.18
3	persistent	middle	335	115	-0.0013	+0.11	-0.0013	-0.12
4	persistent	middle	335	115	-0.0035	+0.20	-0.0026	+0.01
5	persistent	middle	315	118	-0.0012	+0.10	-0.0007	-0.22
3	persistent	high	307	110	+0.0001	-0.05	+0.0007	-0.06
4	persistent	high	294	112	+0.0018	-0.17	+0.0032	-0.12
5	persistent	high	313	112	-0.0018	-0.01	-0.0005	-0.18

Native-prior large runs — TEST excess CE (lower = better; 0 = class prior) per checkpoint (every 2000 steps, checkpoint evals); colour = prior family, line style = evaluation bank; horizontal lines = reference models (test)

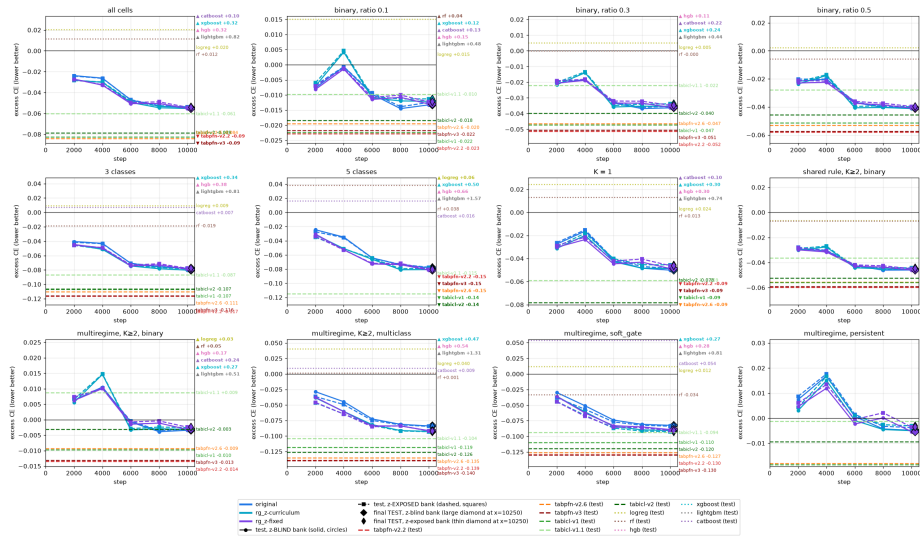


Figure 3: Six-layer native-prior test excess CE by task condition and checkpoint. Solid curves use standard input; dashed curves use duplicate-X1 input (a legacy diagnostic feature, not a true regime ID). Final-checkpoint diamonds are descriptive; matched final comparisons support the headline result.

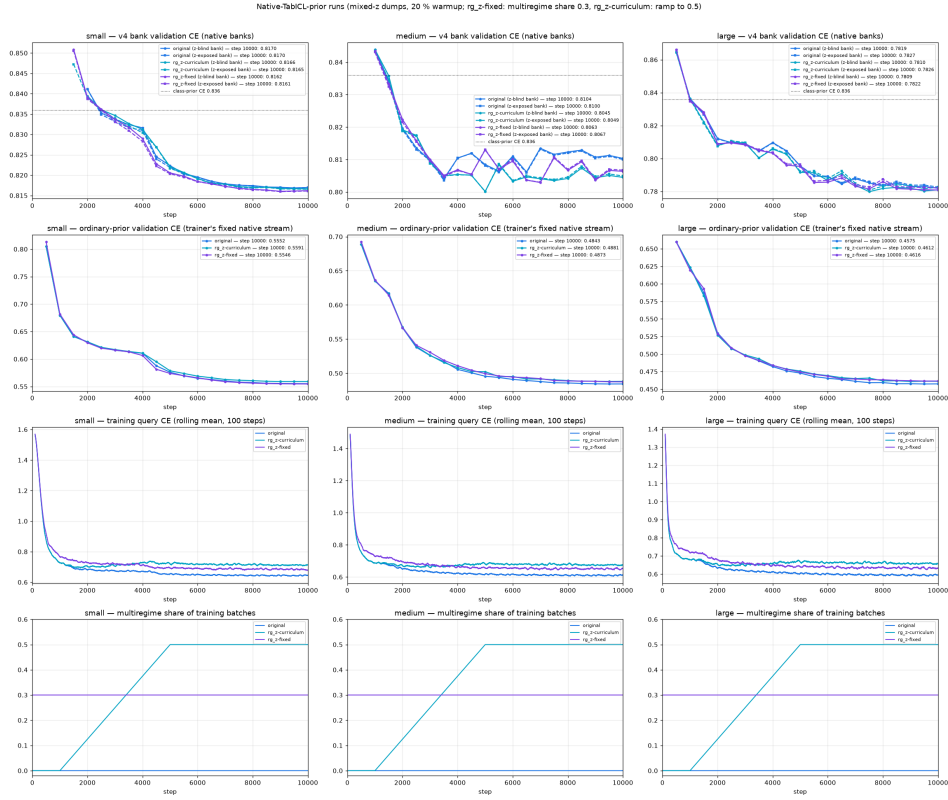


Figure 4: Validation and training trajectories for the three size configurations. Rows show native-bank validation CE, the fixed validation stream, training query CE, and multiregime training-batch share. These diagnostics are distinct from held-out final test comparisons.

D SUPPLIED REGIME-TAG DIAGNOSTIC

Aligned membership as privileged input. A frozen-checkpoint diagnostic evaluates the same 5,760 multiregime multiclass episodes for original, fixed, and curriculum models at all three sizes. The *hidden* condition uses the ordinary features, *shuffled* appends a regime-tag column independently shuffled within support and query, and *true* appends the realized membership. Tag IDs are independently relabelled per episode. True minus shuffled compares aligned membership with a same-width corrupted tag; shuffled minus hidden detects effects from merely appending a feature. The test bank is shared with other analyses, so this is an exploratory diagnostic, not a new untouched test. Section 5 reports the headline true-minus-shuffled and shuffled-minus-hidden ranges; Tables 8–9 below give the full size-by-model contrasts and their paired intervals. Figures 7–9 visualize these input-condition contrasts pooled across and split by SCM mechanism. The 5,000 cell-stratified bootstrap replicates describe finite-bank, not training-seed, uncertainty, and show predictive benefit from a *supplied* aligned tag, not recovery of an unobserved regime.

Table 8: Within-checkpoint cross-entropy contrasts (nats) on multiregime multi-class episodes. True minus shuffled isolates aligned membership at matched input width; shuffled minus hidden captures an appended corrupted tag. Brackets are 95% within-cell episode-bootstrap intervals (5,000 replicates), not training-seed intervals. Each mechanism-by-routing stratum has 1,440 episodes in 180 cells. O/F/C denote original/fixed/curriculum and S/M/L denote size.

Mechanism	Routing	Model/size	True—shuffled [95% interval]	Shuffled—hidden
r_z	persistent	O/S	-0.057 [-0.061, -0.054]	+0.0008
r_z	persistent	F/S	-0.059 [-0.063, -0.055]	+0.0007
r_z	persistent	C/S	-0.058 [-0.061, -0.054]	+0.0008
r_z	persistent	O/M	-0.150 [-0.156, -0.144]	-0.0023
r_z	persistent	F/M	-0.158 [-0.164, -0.152]	-0.0012
r_z	persistent	C/M	-0.164 [-0.170, -0.157]	-0.0018
r_z	persistent	O/L	-0.162 [-0.169, -0.156]	-0.0005
r_z	persistent	F/L	-0.172 [-0.178, -0.165]	-0.0004
r_z	persistent	C/L	-0.174 [-0.180, -0.167]	-0.0005
r_z	soft gate	O/S	-0.035 [-0.037, -0.032]	+0.0048
r_z	soft gate	F/S	-0.035 [-0.037, -0.032]	+0.0046
r_z	soft gate	C/S	-0.034 [-0.037, -0.032]	+0.0048
r_z	soft gate	O/M	-0.062 [-0.065, -0.058]	+0.0057
r_z	soft gate	F/M	-0.057 [-0.060, -0.053]	+0.0046
r_z	soft gate	C/M	-0.054 [-0.057, -0.050]	+0.0037
r_z	soft gate	O/L	-0.042 [-0.046, -0.039]	+0.0044
r_z	soft gate	F/L	-0.038 [-0.042, -0.035]	+0.0025
r_z	soft gate	C/L	-0.037 [-0.040, -0.033]	+0.0024
g_z	persistent	O/S	-0.058 [-0.061, -0.055]	+0.0009
g_z	persistent	F/S	-0.059 [-0.062, -0.056]	+0.0010
g_z	persistent	C/S	-0.058 [-0.061, -0.055]	+0.0011
g_z	persistent	O/M	-0.155 [-0.161, -0.149]	-0.0002
g_z	persistent	F/M	-0.161 [-0.167, -0.155]	-0.0003
g_z	persistent	C/M	-0.167 [-0.173, -0.161]	-0.0008
g_z	persistent	O/L	-0.167 [-0.173, -0.160]	-0.0002
g_z	persistent	F/L	-0.177 [-0.183, -0.170]	-0.0001
g_z	persistent	C/L	-0.179 [-0.185, -0.172]	-0.0001
g_z	soft gate	O/S	-0.035 [-0.038, -0.032]	+0.0049
g_z	soft gate	F/S	-0.035 [-0.037, -0.032]	+0.0047
g_z	soft gate	C/S	-0.034 [-0.037, -0.031]	+0.0048
g_z	soft gate	O/M	-0.062 [-0.066, -0.058]	+0.0045
g_z	soft gate	F/M	-0.058 [-0.062, -0.054]	+0.0039
g_z	soft gate	C/M	-0.056 [-0.059, -0.052]	+0.0033
g_z	soft gate	O/L	-0.044 [-0.048, -0.040]	+0.0041
g_z	soft gate	F/L	-0.040 [-0.044, -0.037]	+0.0026
g_z	soft gate	C/L	-0.038 [-0.042, -0.035]	+0.0022

Table 9: Within-checkpoint accuracy contrasts (percentage points) on multi-regime multiclass episodes. True minus shuffled isolates aligned membership at matched input width; shuffled minus hidden captures an appended corrupted tag. Brackets are 95% within-cell episode-bootstrap intervals (5,000 replicates), not training-seed intervals. Each mechanism-by-routing stratum has 1,440 episodes in 180 cells. O/F/C denote original/fixed/curriculum and S/M/L denote size.

Mechanism	Routing	Model/size	True—shuffled [95% interval]	Shuffled—hidden
r_z	persistent	O/S	+4.52 [+4.18, +4.86]	-0.07
r_z	persistent	F/S	+4.58 [+4.24, +4.93]	-0.03
r_z	persistent	C/S	+4.48 [+4.14, +4.82]	-0.08
r_z	persistent	O/M	+10.15 [+9.72, +10.57]	-0.05
r_z	persistent	F/M	+10.56 [+10.12, +10.99]	-0.07
r_z	persistent	C/M	+10.89 [+10.44, +11.34]	+0.00
r_z	persistent	O/L	+10.85 [+10.41, +11.29]	-0.08
r_z	persistent	F/L	+11.25 [+10.80, +11.69]	+0.01
r_z	persistent	C/L	+11.32 [+10.87, +11.77]	-0.05
r_z	soft gate	O/S	+2.89 [+2.63, +3.17]	-0.39
r_z	soft gate	F/S	+2.69 [+2.41, +2.97]	-0.35
r_z	soft gate	C/S	+2.68 [+2.42, +2.97]	-0.36
r_z	soft gate	O/M	+3.64 [+3.40, +3.89]	-0.36
r_z	soft gate	F/M	+3.18 [+2.94, +3.43]	-0.26
r_z	soft gate	C/M	+3.01 [+2.76, +3.26]	-0.23
r_z	soft gate	O/L	+2.50 [+2.25, +2.75]	-0.22
r_z	soft gate	F/L	+2.18 [+1.95, +2.42]	-0.14
r_z	soft gate	C/L	+1.99 [+1.76, +2.24]	-0.14
g_z	persistent	O/S	+4.62 [+4.29, +4.96]	-0.12
g_z	persistent	F/S	+4.72 [+4.39, +5.07]	-0.10
g_z	persistent	C/S	+4.67 [+4.33, +5.02]	-0.11
g_z	persistent	O/M	+10.67 [+10.23, +11.12]	-0.08
g_z	persistent	F/M	+10.96 [+10.51, +11.41]	+0.00
g_z	persistent	C/M	+11.26 [+10.81, +11.73]	-0.06
g_z	persistent	O/L	+11.20 [+10.71, +11.68]	-0.07
g_z	persistent	F/L	+11.59 [+11.11, +12.07]	-0.09
g_z	persistent	C/L	+11.69 [+11.22, +12.18]	-0.01
g_z	soft gate	O/S	+2.69 [+2.42, +2.98]	-0.27
g_z	soft gate	F/S	+2.62 [+2.35, +2.89]	-0.28
g_z	soft gate	C/S	+2.57 [+2.31, +2.84]	-0.32
g_z	soft gate	O/M	+3.67 [+3.39, +3.95]	-0.27
g_z	soft gate	F/M	+3.41 [+3.12, +3.69]	-0.30
g_z	soft gate	C/M	+3.01 [+2.74, +3.29]	-0.20
g_z	soft gate	O/L	+2.71 [+2.45, +2.98]	-0.31
g_z	soft gate	F/L	+2.35 [+2.08, +2.63]	-0.23
g_z	soft gate	C/L	+2.16 [+1.91, +2.42]	-0.19

What the reports cannot show. The available full-bank reports record episode-level CE and accuracy by input condition, rather than query-level predictions paired with each query’s true z . They cannot estimate query-level regime recovery or per-regime error. A per-regime analysis would require those paired predictions and within-episode label alignment; arbitrary episode-local regime numbers cannot be pooled directly. Predictive accuracy under a supplied tag would still not establish latent-regime identification (Jiang & Tanner, 1999).

Checkpoint provenance and interpretation. The nine diagnostic report records point to the ordinary native `original`, `rg_z-fixed`, and `rg_z-curriculum` final checkpoints, not to a distinct set of checkpoints trained with true row-level membership. The report records also have different evaluator model-source hashes for original and candidate loads; an isolated effect of that code-path difference has not been established. We therefore treat cross-model contrasts in Figures 5 and 6 as descriptive, and do not interpret them as effects of true-tag training or a particular attention computation. Within a frozen checkpoint, true versus shuffled tags compare aligned information at

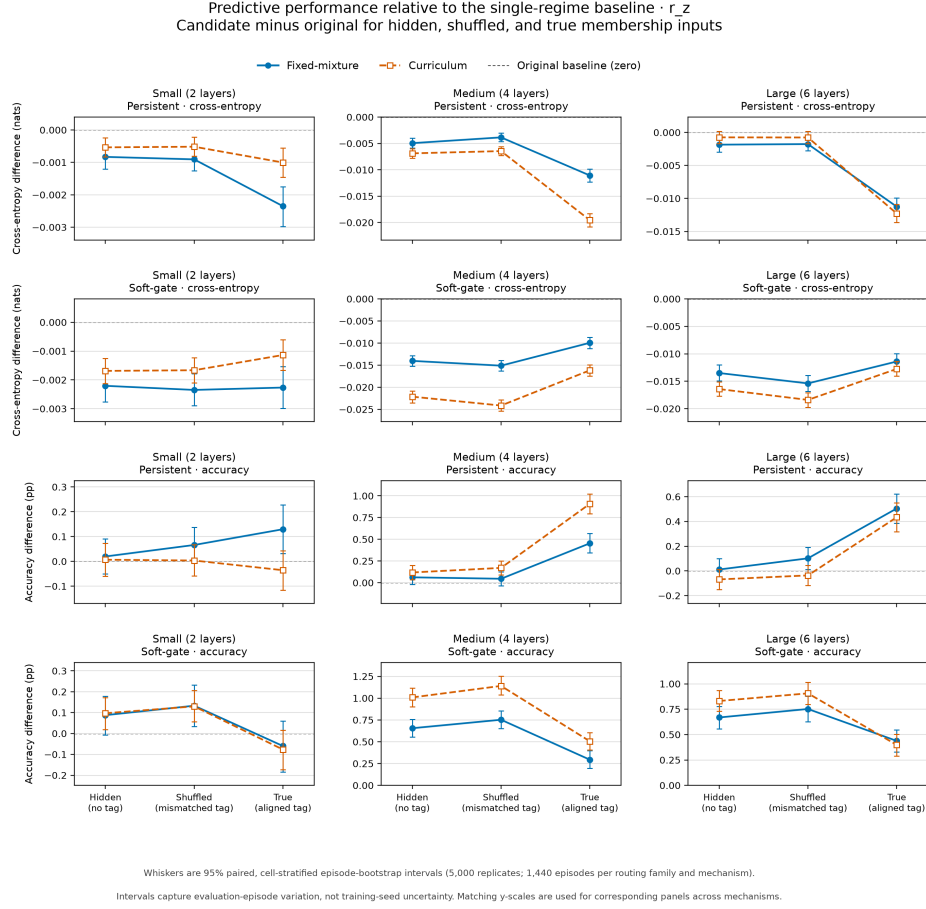


Figure 5: Frozen native-model differences from the original control for r_z episodes under hidden, shuffled, and true membership inputs. Rows separate routing families and metrics; columns separate model sizes. Whiskers show paired within-cell episode-bootstrap intervals. The reports do not identify separately trained true-tag checkpoints.

the same input width. A separate, much smaller monitoring probe used during training (42 episodes) is validation material and is not treated as final-test evidence.

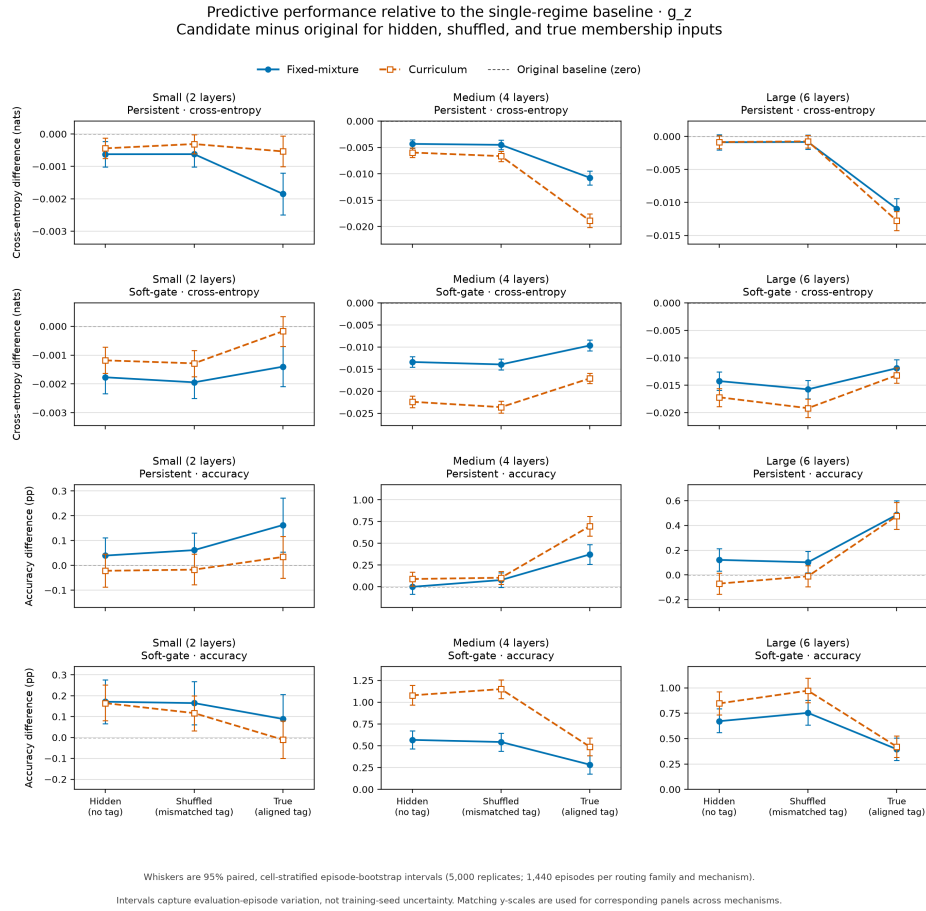


Figure 6: As in Figure 5, for g_z episodes. The within-model true-minus-shuffled contrast is the relevant supplied-information comparison.

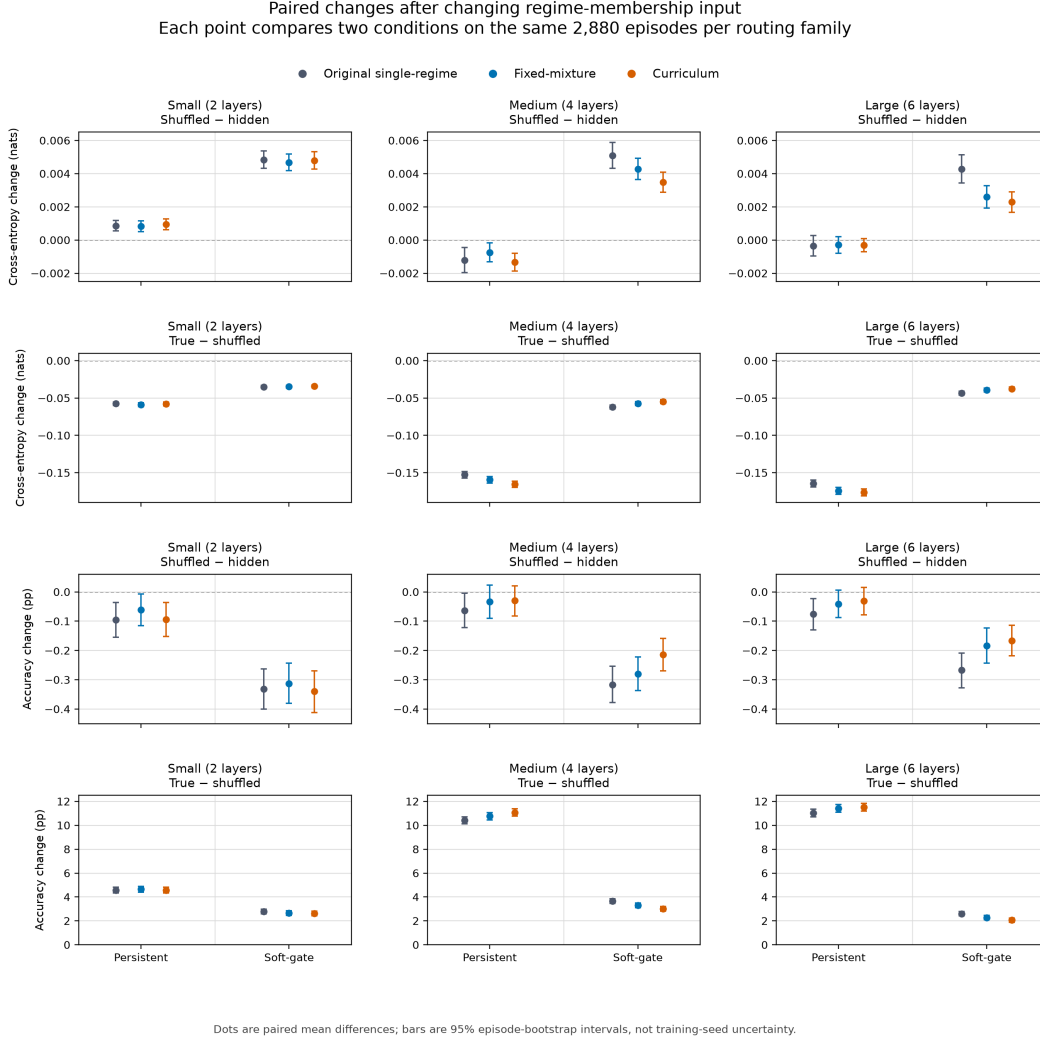
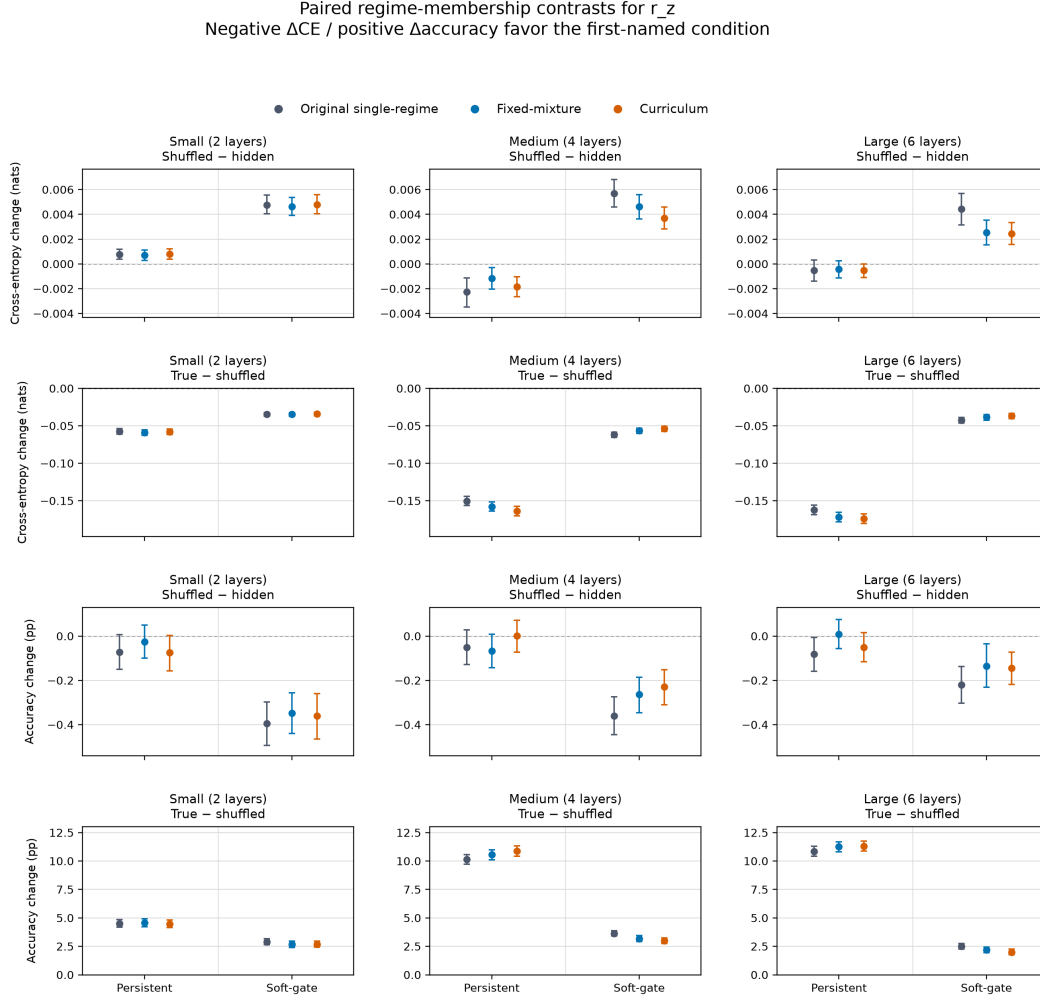
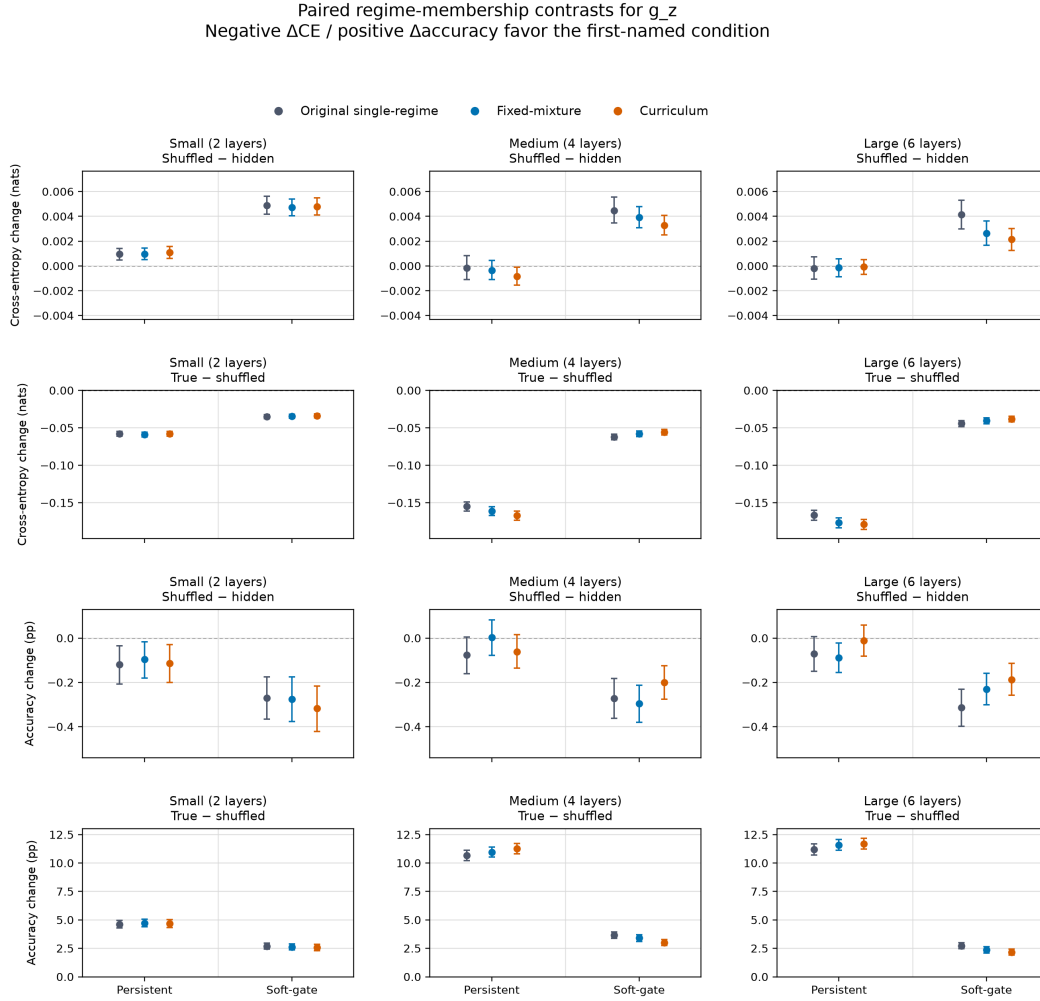


Figure 7: Paired input-condition contrasts pooled across the r_z and g_z mechanisms. Columns show model size; rows show cross-entropy and accuracy changes for shuffled versus hidden and true versus shuffled inputs under persistent and soft-gate routing. Each routing family includes 2,880 matched episodes. Dots are paired mean differences; bars are 95% cell-stratified episode-bootstrap intervals over this fixed bank, not training-seed uncertainty. Negative cross-entropy and positive accuracy changes favor the first-named condition.



N=1,440 matched episodes per routing family and mechanism. Dots are paired mean differences; bars are 95% cell-stratified episode-bootstrap intervals, not training-seed uncertainty.

Figure 8: Within-model paired input-condition contrasts for r_z episodes. Panels show cross-entropy and accuracy changes for shuffled versus hidden and true versus shuffled inputs across sizes and routing families. Each routing family includes 1,440 matched episodes. Dots are paired means and bars are 95% cell-stratified episode-bootstrap intervals over this fixed bank, not training-seed uncertainty. Negative cross-entropy and positive accuracy changes favor the first-named condition.



N=1,440 matched episodes per routing family and mechanism. Dots are paired mean differences; bars are 95% cell-stratified episode-bootstrap intervals, not training-seed uncertainty.

Figure 9: As in Figure 8, for g_z episodes. Each routing family has 1,440 matched episodes; intervals are cell-stratified episode-bootstrap intervals over the fixed bank, not training-seed uncertainty.

E TRAINING-TIME EXPOSURE TO REGIME IDS

This exploratory comparison uses the saved two-layer (small) and four-layer (medium) checkpoints trained with row-level regime-ID exposure enabled (seed 2402, step 10,000). It compares each with the same-size original, fixed-mixture, and curriculum checkpoints. This is a training-time intervention: the IDs are supplied while the observed-ID variants are trained. It is distinct from Section D, which keeps checkpoints frozen and varies whether an ID is supplied only at evaluation. The hidden-input condition is the primary setting because the prediction task does not provide a regime ID; shuffled-tag and true-tag scores are shown as context.

All eight archived reports use the same recorded bank path and contain identical episode IDs and episode metadata for the 5,760 multiclass multiregime episodes (720 cells, eight episodes per cell). Table 10 reports cross-entropy and accuracy for hidden, shuffled-tag, and true-tag inputs. The observed-minus-original hidden contrasts pair episodes and average cell means equally. Their 95% intervals resample episodes within cells and quantify uncertainty over this fixed episode bank only, not variation across training seeds.

The reports record different model-source hashes across model families, do not record evaluator-source hashes, and do not include a content hash for the bank file. We use the archived full-bank scores; the reports’ shared path, IDs, and metadata establish alignment but do not verify the bank file’s byte-level identity. Accordingly, this single-seed comparison is descriptive and does not isolate a causal effect of training with regime IDs.

Table 10: Exploratory observed-ID training comparison on 5,760 multiclass multiregime episodes. H = hidden, S = shuffled-tag, and T = true-tag evaluation inputs; the hidden condition is the primary task setting. Cross-model observed-minus-original differences are paired by episode under hidden inputs. Brackets give 95% within-cell episode-bootstrap intervals (5,000 replicates); they quantify this fixed episode bank and not pretraining-seed variation. The saved model-source hashes differ across model families, so the cross-model comparison is descriptive.

Size	Model	Cross-entropy (nats)			Accuracy (%)			Observed-ID — original, hidden [95% interval]	
		H	S	T	H	S	T	CE	Accuracy (pp)
Small	Original	1.221	1.224	1.177	44.1	43.9	47.6	—	—
Small	Fixed mixture	1.219	1.222	1.175	44.2	44.0	47.7	—	—
Small	Curriculum	1.220	1.223	1.177	44.2	44.0	47.6	—	—
Small	Observed-ID training	1.226	1.226	1.183	43.8	43.8	47.3	+0.005 [+0.004, +0.006]	-0.33 [-0.42, -0.24]
Medium	Original	1.192	1.194	1.087	47.1	46.9	53.9	—	—
Medium	Fixed mixture	1.183	1.185	1.076	47.4	47.2	54.2	—	—
Medium	Curriculum	1.178	1.179	1.069	47.6	47.5	54.5	—	—
Medium	Observed-ID training	1.194	1.195	1.096	46.2	46.1	52.9	+0.002 [+0.001, +0.004]	-0.87 [-0.96, -0.77]

F BEYONDARENA TASK INVENTORY AND RESULTS

This is a selected 36-task subset of BeyondArena (Purucker et al., 2026): 26 binary IID, one binary grouped, six multiclass IID, two multiclass grouped, and one multiclass temporal task. The three grouped tasks hold out many small groups rather than a few large ones, computed directly from each task’s group column via Data Foundry 0.0.5: parkinsons_biomedical_voice_measurements has 32 patient_id groups (195 rows, mean 6.1 rows/group, range 6–7); cardiocography has 176 patient_id groups (2,126 rows, mean 12.1, range 1–54); and dementia_prediction has 150 Subject ID groups (370 rows, mean 2.5, range 1–5). The evaluator uses Data Foundry containers and official outer folds, fits the feature preprocessor on each training fold only, passes the complete training fold as PFN context, and scores the official query fold. Its configured filter allows at most 11,000 rows, 30 raw features, and 2–5 classes; other eligibility checks exclude text and high-cardinality features. The accompanying beyondarena_36_tasks.csv resolves all result-table names to container UUIDs and checksums, and beyondarena_36_official_folds.csv archives 1,440 official repeat/fold identifiers, fold sizes, and train/test index hashes. These files were reconstructed with Data Foundry 0.0.5 from immutable BeyondArena revision 2ecfe882ccfb814fc27c4de10a64ceefd5d7655c. All 36 container checksums match the saved inventory; the 32 tasks evaluated in the earlier local run also match its recorded fold IDs and sizes. The original 36-task job did not log its cache commit, so exact historical revision identity remains unverified. Task scores average folds, and aggregate scores weight tasks equally. The 35-task native-rank graphic excludes the one temporal task and ranks nine native runs only; the 36-task score tables include that task and published/conventional comparators. Ranks over different comparison sets must not be mixed (Demšar, 2006).

Table 11: BeyondArena equal-task summaries from the original evaluation aggregates. Overall excess CE covers 36 tasks, binary accuracy gain 27 tasks, and multiclass accuracy gain nine tasks. Values are rounded; accuracy gains are percentage points.

Model	Overall excess CE ↓	Binary gain	Multiclass gain
Native single-regime	-.224	9.4	20.0
Native fixed mixture	-.221	9.3	20.2
Native curriculum	-.220	9.2	19.9
Random forest	-.225	9.7	21.7
TabICL v2	-.273	11.2	23.0
TabPFN v3	-.267	11.0	22.8

Table 12: BeyondArena split summaries, Excess CE (nats; lower is better). Equal-task means are calculated from task-level results rounded to three decimals and are descriptive. Grouped and temporal slices contain only one or two tasks each.

Model	Bin. IID (26)	Bin. group (1)	Multi IID (6)	Multi group (2)	Multi time (1)
Native single-regime	-0.161	+0.067	-0.525	-0.346	-0.095
Native fixed mixture	-0.159	+0.089	-0.527	-0.347	-0.069
Native curriculum	-0.159	+0.127	-0.526	-0.348	-0.088
Random forest	-0.160	-0.118	-0.567	-0.345	+0.267
TabICL v2	-0.206	+0.116	-0.620	-0.360	-0.139
TabPFN v3	-0.205	+0.260	-0.616	-0.345	-0.142

Table 13: BeyondArena split summaries, Accuracy gain (percentage points; higher is better). Equal-task means are calculated from task-level results rounded to three decimals and are descriptive. Grouped and temporal slices contain only one or two tasks each.

Model	Bin. IID (26)	Bin. group (1)	Multi IID (6)	Multi group (2)	Multi time (1)
Native single-regime	+9.55	+4.10	+25.78	+12.80	-0.80
Native fixed mixture	+9.47	+4.50	+26.03	+13.05	-0.90
Native curriculum	+9.45	+3.80	+25.87	+12.55	-1.40
Random forest	+9.88	+3.90	+28.47	+12.35	-0.40
TabICL v2	+11.45	+3.40	+30.07	+13.35	-0.30
TabPFN v3	+11.41	+1.00	+30.05	+12.80	-0.30

Table 14: Selected BeyondArena task inventory. Training rows and features describe the task-level evaluation manifest; G = grouped and T = temporal. Task names follow the source results and are truncated there where necessary.

Task	Split	Train n	d	Majority
bad_customer_detection	IID	1148	13	0.89
bank_customer_churn	IID	6666	10	0.80
blood_transfusion	IID	498	4	0.76
churn	IID	3333	19	0.86
credit_approval	IID	460	15	0.56
credit_g	IID	666	20	0.70
early_stage_diabetes_risk_pred	IID	167	16	0.69
ecommerce_shipping	IID	7332	10	0.60
fitness_club	IID	1000	6	0.70
hazelnut_spread_contaminant_de	IID	1600	30	0.50
heart_disease_cleveland	IID	202	13	0.54
heart_disease_hungary	IID	196	13	0.64
heart_disease_va_long_beach	IID	133	13	0.74
heart_failure_followup_surviva	IID	199	12	0.68
heloc	IID	6972	23	0.52
hepatitis_survival_prediction	IID	103	19	0.79
homeq_default_prediction	IID	3805	12	0.80
indian_liver_patient_dataset	IID	388	10	0.71
iranian_churn	IID	1900	13	0.84
jm1	IID	7256	21	0.81
ljublana_breast_cancer	IID	190	9	0.70
marketing_campaign	IID	1493	25	0.85
seismic_bumps	IID	1722	15	0.93
south_africa_coronary_heart_di	IID	308	9	0.65
thyroid_discordant	IID	2474	26	0.98
tour_travels_churn	IID	636	6	0.77
parkinsons_biomedical_voice_me	G	127	23	0.76
biomechanical_orthopaedic_pred	IID	206	6	0.48

Task	Split	Train n	d	Majority
ecoli_proteins	IID	218	6	0.44
hepatitis_c_prediction	IID	405	12	0.88
horse_colic_survival	IID	229	20	0.63
maternal_health_risk	IID	676	6	0.40
website_phishing	IID	902	9	0.52
cardiotocography	G	1254	22	0.78
dementia_prediction	G	250	8	0.56
ghanas_indigenous_intel	T	9933	10	0.93

Table 15: Per-task BeyondArena Excess cross-entropy (nats; lower is better): small/medium native models. O/F/C are native single-regime/fixed/curriculum; S/M/L indicate size. Results are rounded task aggregates.

Task	Split	O-S	F-S	C-S	O-M	F-M	C-M
bad_customer_detection	IID	-0.019	-0.018	-0.015	-0.028	-0.025	-0.026
bank_customer_churn	IID	-0.075	-0.079	-0.080	-0.137	-0.133	-0.129
blood_transfusion	IID	-0.074	-0.072	-0.076	-0.065	-0.067	-0.064
churn	IID	-0.076	-0.075	-0.067	-0.136	-0.131	-0.133
credit_approval	IID	-0.308	-0.316	-0.313	-0.347	-0.342	-0.340
credit_g	IID	-0.075	-0.072	-0.075	-0.099	-0.092	-0.091
early_stage_diabetes_risk_pred	IID	-0.297	-0.288	-0.285	-0.349	-0.351	-0.337
ecommerce_shipping	IID	-0.124	-0.130	-0.129	-0.154	-0.157	-0.154
fitness_club	IID	-0.118	-0.122	-0.123	-0.139	-0.140	-0.141
hazelnut_spread_contaminant_de	IID	-0.220	-0.225	-0.227	-0.314	-0.309	-0.305
heart_disease_cleveland	IID	-0.261	-0.259	-0.259	-0.287	-0.289	-0.287
heart_disease_hungary	IID	-0.241	-0.245	-0.242	-0.286	-0.288	-0.294
heart_disease_va_long_beach	IID	-0.028	-0.026	-0.031	+0.000	-0.000	-0.013
heart_failure_followup_surviva	IID	-0.208	-0.197	-0.181	-0.261	-0.263	-0.264
heloc	IID	-0.106	-0.116	-0.117	-0.129	-0.132	-0.128
hepatitis_survival_prediction	IID	-0.142	-0.137	-0.147	-0.118	-0.125	-0.129
homeq_default_prediction	IID	-0.141	-0.150	-0.145	-0.205	-0.207	-0.190
indian_liver_patient_dataset	IID	-0.059	-0.064	-0.065	-0.075	-0.078	-0.081
iranian_churn	IID	-0.183	-0.190	-0.185	-0.230	-0.223	-0.220
jm1	IID	+0.007	-0.019	-0.010	-0.039	-0.038	-0.037
ljubljana_breast_cancer	IID	-0.043	-0.045	-0.048	-0.050	-0.052	-0.049
marketing_campaign	IID	-0.079	-0.074	-0.073	-0.124	-0.112	-0.113
seismic_bumps	IID	-0.022	-0.022	-0.023	-0.030	-0.029	-0.030
south_africa_coronary_heart_di	IID	-0.083	-0.079	-0.081	-0.090	-0.091	-0.091
thyroid_discordant	IID	+0.082	+0.043	+0.034	-0.014	-0.011	-0.010
tour_travels_churn	IID	-0.123	-0.120	-0.118	-0.211	-0.201	-0.192
parkinsons_biomedical_voice_me	G	-0.075	-0.070	-0.076	+0.004	-0.004	+0.009
biomechanical_orthopaedic_pred	IID	-0.406	-0.409	-0.407	-0.627	-0.637	-0.641
ecoli_proteins	IID	-0.563	-0.560	-0.548	-0.879	-0.839	-0.855
hepatitis_c_prediction	IID	-0.168	-0.187	-0.158	-0.341	-0.333	-0.330
horse_colic_survival	IID	-0.036	-0.017	-0.013	-0.205	-0.201	-0.202
maternal_health_risk	IID	-0.256	-0.246	-0.239	-0.412	-0.414	-0.419
website_phishing	IID	-0.272	-0.270	-0.271	-0.489	-0.473	-0.474
cardiotocography	G	-0.212	-0.224	-0.209	-0.359	-0.369	-0.352
dementia_prediction	G	-0.228	-0.249	-0.236	-0.282	-0.303	-0.293
ghanas_indigenous_intel	T	-0.072	-0.079	-0.068	-0.115	-0.105	-0.099

Table 16: Per-task BeyondArena Excess cross-entropy (nats; lower is better): large native and reference models. O/F/C are native single-regime/fixed/curriculum; S/M/L indicate size. Results are rounded task aggregates.

Task	Split	O-L	F-L	C-L	PFN3	ICL2	RF	LR
bad_customer_detection	IID	-0.037	-0.038	-0.036	-0.038	-0.037	-0.030	-0.025
bank_customer_churn	IID	-0.152	-0.152	-0.150	-0.180	-0.176	-0.156	-0.069
blood_transfusion	IID	-0.067	-0.069	-0.068	-0.072	-0.074	+0.116	-0.069
churn	IID	-0.166	-0.161	-0.161	-0.294	-0.286	-0.224	-0.087
credit_approval	IID	-0.357	-0.356	-0.325	-0.371	-0.374	-0.346	-0.334
credit_g	IID	-0.103	-0.102	-0.100	-0.117	-0.121	-0.109	-0.067
early_stage_diabetes_risk_pred	IID	-0.349	-0.346	-0.347	-0.404	-0.428	-0.387	-0.336
ecommerce_shipping	IID	-0.162	-0.163	-0.164	-0.178	-0.177	-0.163	-0.127
fitness_club	IID	-0.140	-0.142	-0.140	-0.147	-0.148	-0.088	-0.136
hazelnut_spread_contaminant_de	IID	-0.387	-0.378	-0.377	-0.604	-0.610	-0.407	-0.421
heart_disease_cleveland	IID	-0.268	-0.264	-0.272	-0.301	-0.300	-0.285	-0.269
heart_disease_hungary	IID	-0.264	-0.265	-0.274	-0.283	-0.285	-0.244	-0.274
heart_disease_va_long_beach	IID	-0.011	-0.003	+0.001	-0.019	-0.008	-0.024	+0.059
heart_failure_followup_surviva	IID	-0.252	-0.238	-0.254	-0.264	-0.274	-0.262	-0.198
heloc	IID	-0.140	-0.135	-0.140	-0.150	-0.149	-0.139	-0.127
hepatitis_survival_prediction	IID	-0.116	-0.107	-0.119	-0.131	-0.148	-0.168	-0.006
homeq_default_prediction	IID	-0.231	-0.229	-0.224	-0.468	-0.432	-0.293	-0.220
indian_liver_patient_dataset	IID	-0.079	-0.082	-0.083	-0.098	-0.100	-0.079	-0.081
iranian_churn	IID	-0.263	-0.257	-0.252	-0.386	-0.385	-0.310	-0.210
jm1	IID	-0.045	-0.046	-0.047	-0.078	-0.076	-0.034	-0.044
ljublana_breast_cancer	IID	-0.051	-0.049	-0.055	-0.055	-0.057	+0.025	-0.022
marketing_campaign	IID	-0.151	-0.141	-0.143	-0.171	-0.189	-0.142	-0.134
seismic_bumps	IID	-0.031	-0.033	-0.032	-0.032	-0.032	-0.001	-0.025
south_africa_coronary_heart_di	IID	-0.086	-0.087	-0.087	-0.098	-0.103	-0.069	-0.105
thyroid_discordant	IID	-0.025	-0.026	-0.023	-0.053	-0.054	-0.044	-0.010
tour_travels_churn	IID	-0.263	-0.259	-0.252	-0.346	-0.341	-0.294	-0.134
parkinsons_biomedical_voice_me	G	+0.067	+0.089	+0.127	+0.260	+0.116	-0.118	+0.041
biomechanical_orthopaedic_pred	IID	-0.669	-0.673	-0.674	-0.701	-0.701	-0.667	-0.680
ecoli_proteins	IID	-0.892	-0.895	-0.898	-1.038	-1.040	-0.993	-1.010
hepatitis_c_prediction	IID	-0.355	-0.370	-0.362	-0.385	-0.385	-0.326	-0.285
horse_colic_survival	IID	-0.191	-0.191	-0.198	-0.189	-0.204	-0.183	+0.091
maternal_health_risk	IID	-0.487	-0.483	-0.480	-0.701	-0.709	-0.639	-0.298
website_phishing	IID	-0.554	-0.551	-0.542	-0.684	-0.681	-0.593	-0.396
cardiotocography	G	-0.403	-0.404	-0.399	-0.439	-0.451	-0.436	-0.334
dementia_prediction	G	-0.289	-0.289	-0.297	-0.251	-0.269	-0.253	-0.337
ghanas_indigenous_intel	T	-0.095	-0.069	-0.088	-0.142	-0.139	+0.267	-0.081

Table 17: Per-task BeyondArena Accuracy gain (fraction; higher is better): small/medium native models. O/F/C are native single-regime/fixed/curriculum; S/M/L indicate size. Results are rounded task aggregates.

Task	Split	O-S	F-S	C-S	O-M	F-M	C-M
bad_customer_detection	IID	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000
bank_customer_churn	IID	+0.011	+0.011	+0.011	+0.054	+0.048	+0.050
blood_transfusion	IID	+0.019	+0.009	+0.021	+0.011	+0.014	+0.016

Continued from previous page

Task	Split	O-S	F-S	C-S	O-M	F-M	C-M
churn	IID	-0.006	-0.005	-0.004	+0.024	+0.022	+0.024
credit_approval	IID	+0.287	+0.289	+0.285	+0.305	+0.300	+0.303
credit_g	IID	+0.016	+0.016	+0.020	+0.040	+0.034	+0.037
early_stage_diabetes_risk_pred	IID	+0.180	+0.172	+0.170	+0.181	+0.178	+0.169
ecommerce_shipping	IID	+0.062	+0.069	+0.073	+0.076	+0.081	+0.077
fitness_club	IID	+0.076	+0.075	+0.076	+0.073	+0.073	+0.077
hazelnut_spread_contaminant_de	IID	+0.282	+0.292	+0.294	+0.337	+0.340	+0.334
heart_disease_cleveland	IID	+0.265	+0.258	+0.254	+0.281	+0.279	+0.281
heart_disease_hungary	IID	+0.186	+0.187	+0.181	+0.204	+0.199	+0.198
heart_disease_va_long_beach	IID	+0.007	+0.003	+0.012	-0.000	+0.003	+0.005
heart_failure_followup_surviva	IID	+0.136	+0.123	+0.103	+0.153	+0.153	+0.161
heloc	IID	+0.177	+0.183	+0.184	+0.195	+0.197	+0.195
hepatitis_survival_prediction	IID	+0.034	+0.036	+0.038	+0.049	+0.048	+0.043
homeq_default_prediction	IID	+0.044	+0.052	+0.052	+0.078	+0.077	+0.067
indian_liver_patient_dataset	IID	-0.030	-0.027	-0.025	+0.001	-0.003	-0.001
iranian_churn	IID	+0.044	+0.049	+0.048	+0.068	+0.071	+0.067
jml	IID	-0.050	-0.013	-0.031	+0.001	+0.002	+0.002
ljublana_breast_cancer	IID	+0.005	+0.004	+0.005	+0.020	+0.024	+0.018
marketing_campaign	IID	+0.009	+0.013	+0.014	+0.024	+0.020	+0.020
seismic_bumps	IID	+0.000	+0.000	+0.000	+0.000	+0.000	-0.000
south_africa_coronary_heart_di	IID	+0.053	+0.045	+0.045	+0.058	+0.057	+0.059
thyroid_discordant	IID	-0.044	-0.000	+0.000	+0.000	+0.000	-0.001
tour_travels_churn	IID	+0.044	+0.041	+0.033	+0.058	+0.059	+0.054
parkinsons_biomedical_voice_me	G	-0.025	-0.034	-0.021	+0.039	+0.043	+0.044
biomechanical_orthopaedic_pred	IID	+0.233	+0.237	+0.229	+0.351	+0.342	+0.350
ecoli_proteins	IID	+0.210	+0.202	+0.204	+0.388	+0.362	+0.375
hepatitis_c_prediction	IID	+0.006	+0.009	+0.003	+0.059	+0.057	+0.056
horse_colic_survival	IID	+0.002	-0.000	+0.004	+0.054	+0.057	+0.061
maternal_health_risk	IID	+0.202	+0.192	+0.194	+0.280	+0.283	+0.294
website_phishing	IID	+0.247	+0.251	+0.245	+0.313	+0.314	+0.310
cardiotocography	G	+0.023	+0.025	+0.024	+0.077	+0.085	+0.071
dementia_prediction	G	+0.123	+0.148	+0.137	+0.171	+0.167	+0.167
ghanas_indigenous_intel	T	+0.000	+0.000	+0.000	-0.015	-0.011	-0.003

Table 18: Per-task BeyondArena Accuracy gain (fraction; higher is better): large native and reference models. O/F/C are native single-regime/fixed/curriculum; S/M/L indicate size. Results are rounded task aggregates.

Task	Split	O-L	F-L	C-L	PFN3	ICL2	RF	LR
bad_customer_detection	IID	-0.000	+0.000	+0.000	-0.001	-0.001	-0.001	-0.001
bank_customer_churn	IID	+0.058	+0.057	+0.058	+0.071	+0.069	+0.062	+0.011
blood_transfusion	IID	+0.023	+0.034	+0.033	+0.025	+0.029	-0.018	+0.011
churn	IID	+0.048	+0.045	+0.044	+0.114	+0.111	+0.098	+0.007
credit_approval	IID	+0.315	+0.315	+0.286	+0.315	+0.315	+0.322	+0.307
credit_g	IID	+0.050	+0.049	+0.051	+0.061	+0.061	+0.054	+0.018
early_stage_diabetes_risk_pred	IID	+0.198	+0.191	+0.191	+0.215	+0.231	+0.231	+0.175
ecommerce_shipping	IID	+0.076	+0.076	+0.083	+0.089	+0.088	+0.070	+0.042
fitness_club	IID	+0.074	+0.076	+0.078	+0.084	+0.084	+0.056	+0.079

Continued from previous page

Task	Split	O-L	F-L	C-L	PFN3	ICL2	RF	LR
hazelnut_spread_contaminant_de	IID	+0.369	+0.364	+0.362	+0.468	+0.470	+0.393	+0.389
heart_disease_cleveland	IID	+0.273	+0.273	+0.283	+0.292	+0.292	+0.280	+0.281
heart_disease_hungary	IID	+0.202	+0.203	+0.204	+0.197	+0.195	+0.178	+0.201
heart_disease_va_long_beach	IID	+0.011	+0.008	+0.006	+0.001	-0.002	+0.006	-0.014
heart_failure_followup_surviva	IID	+0.147	+0.145	+0.154	+0.165	+0.172	+0.165	+0.140
heloc	IID	+0.198	+0.199	+0.199	+0.210	+0.210	+0.201	+0.194
hepatitis_survival_prediction	IID	+0.047	+0.043	+0.042	+0.041	+0.043	+0.056	+0.033
homeq_default_prediction	IID	+0.096	+0.101	+0.099	+0.189	+0.174	+0.111	+0.091
indian_liver_patient_dataset	IID	+0.001	-0.008	-0.006	+0.005	+0.004	-0.003	+0.003
iranian_churn	IID	+0.082	+0.076	+0.071	+0.138	+0.138	+0.112	+0.054
jm1	IID	+0.004	+0.003	+0.005	+0.017	+0.014	+0.011	+0.006
ljublana_breast_cancer	IID	+0.034	+0.036	+0.039	+0.033	+0.036	+0.005	+0.017
marketing_campaign	IID	+0.038	+0.033	+0.035	+0.048	+0.046	+0.030	+0.033
seismic_bumps	IID	+0.000	+0.000	+0.000	-0.000	-0.000	-0.002	-0.003
south_africa_coronary_heart_di	IID	+0.053	+0.058	+0.058	+0.064	+0.071	+0.037	+0.069
thyroid_discordant	IID	+0.000	+0.000	+0.000	+0.006	+0.007	+0.002	-0.002
tour_travels_churn	IID	+0.086	+0.086	+0.082	+0.120	+0.120	+0.112	+0.054
parkinsons_biomedical_voice_me	G	+0.041	+0.045	+0.038	+0.010	+0.034	+0.039	+0.024
biomechanical_orthopaedic_pred	IID	+0.364	+0.371	+0.367	+0.377	+0.375	+0.360	+0.368
ecoli_proteins	IID	+0.396	+0.392	+0.392	+0.444	+0.443	+0.440	+0.447
hepatitis_c_prediction	IID	+0.065	+0.071	+0.068	+0.080	+0.081	+0.059	+0.063
horse_colic_survival	IID	+0.052	+0.060	+0.065	+0.060	+0.058	+0.053	+0.019
maternal_health_risk	IID	+0.326	+0.322	+0.321	+0.445	+0.450	+0.425	+0.216
website_phishing	IID	+0.344	+0.346	+0.339	+0.397	+0.397	+0.371	+0.305
cardiotocography	G	+0.101	+0.103	+0.096	+0.129	+0.132	+0.126	+0.090
dementia_prediction	G	+0.155	+0.158	+0.155	+0.127	+0.135	+0.121	+0.170
ghanas_indigenous_intel	T	-0.008	-0.009	-0.014	-0.003	-0.003	-0.004	-0.026

Table 19: Per-task BeyondArena Macro one-versus-rest AUC (higher is better): small/medium native models. O/F/C are native single-regime/fixed/curriculum; S/M/L indicate size. Results are rounded task aggregates.

Task	Split	O-S	F-S	C-S	O-M	F-M	C-M
bad_customer_detection	IID	0.693	0.689	0.678	0.736	0.734	0.735
bank_customer_churn	IID	0.787	0.791	0.790	0.843	0.841	0.836
blood_transfusion	IID	0.755	0.757	0.758	0.739	0.741	0.735
churn	IID	0.841	0.833	0.825	0.880	0.875	0.878
credit_approval	IID	0.907	0.910	0.908	0.925	0.924	0.923
credit_g	IID	0.743	0.741	0.745	0.777	0.774	0.771
early_stage_diabetes_risk_pred	IID	0.942	0.932	0.930	0.952	0.953	0.947
ecommerce_shipping	IID	0.731	0.734	0.734	0.740	0.740	0.740
fitness_club	IID	0.786	0.793	0.793	0.813	0.816	0.817
hazelnut_spread_contaminant_de	IID	0.877	0.877	0.877	0.913	0.910	0.910
heart_disease_cleveland	IID	0.886	0.882	0.882	0.901	0.901	0.897
heart_disease_hungary	IID	0.884	0.884	0.884	0.911	0.912	0.913
heart_disease_va_long_beach	IID	0.682	0.679	0.684	0.659	0.652	0.663
heart_failure_followup_surviva	IID	0.883	0.875	0.864	0.910	0.911	0.910
heloc	IID	0.762	0.773	0.774	0.784	0.785	0.781

Continued from previous page

Task	Split	O-S	F-S	C-S	O-M	F-M	C-M
hepatitis_survival_prediction	IID	0.869	0.857	0.862	0.863	0.868	0.857
homeq_default_prediction	IID	0.846	0.858	0.853	0.905	0.909	0.898
indian_liver_patient_dataset	IID	0.714	0.715	0.718	0.731	0.735	0.737
iranian_churn	IID	0.915	0.914	0.914	0.954	0.950	0.948
jm1	IID	0.691	0.688	0.689	0.706	0.708	0.706
ljubljana_breast_cancer	IID	0.715	0.717	0.720	0.709	0.711	0.709
marketing_campaign	IID	0.811	0.807	0.803	0.874	0.864	0.864
seismic_bumps	IID	0.746	0.749	0.749	0.770	0.766	0.770
south_africa_coronary_heart_di	IID	0.746	0.741	0.743	0.758	0.759	0.759
thyroid_discordant	IID	0.580	0.547	0.543	0.857	0.837	0.843
tour_travels_churn	IID	0.828	0.825	0.824	0.897	0.891	0.887
parkinsons_biomedical_voice_me	G	0.818	0.809	0.811	0.778	0.776	0.783
biomechanical_orthopaedic_pred	IID	0.874	0.879	0.883	0.945	0.946	0.947
ecoli_proteins	IID	0.859	0.865	0.866	0.955	0.943	0.950
hepatitis_c_prediction	IID	0.917	0.921	0.914	0.972	0.974	0.972
horse_colic_survival	IID	0.719	0.718	0.724	0.805	0.800	0.803
maternal_health_risk	IID	0.781	0.775	0.773	0.850	0.852	0.853
website_phishing	IID	0.833	0.830	0.829	0.924	0.921	0.917
cardiotocography	G	0.880	0.881	0.868	0.948	0.950	0.948
dementia_prediction	G	0.825	0.822	0.823	0.851	0.853	0.849
ghanas_indigenous_intel	T	0.857	0.858	0.852	0.875	0.877	0.873

Table 20: Per-task BeyondArena Macro one-versus-rest AUC (higher is better); large native and reference models. O/F/C are native single-regime/fixed/curriculum; S/M/L indicate size. Results are rounded task aggregates.

Task	Split	O-L	F-L	C-L	PFN3	ICL2	RF	LR
bad_customer_detection	IID	0.745	0.751	0.745	0.747	0.746	0.728	0.705
bank_customer_churn	IID	0.848	0.850	0.847	0.875	0.871	0.851	0.753
blood_transfusion	IID	0.746	0.742	0.739	0.753	0.753	0.682	0.751
churn	IID	0.894	0.890	0.891	0.933	0.931	0.916	0.823
credit_approval	IID	0.933	0.933	0.931	0.937	0.939	0.934	0.920
credit_g	IID	0.780	0.778	0.778	0.793	0.796	0.784	0.729
early_stage_diabetes_risk_pred	IID	0.956	0.952	0.953	0.969	0.975	0.973	0.945
ecommerce_shipping	IID	0.743	0.738	0.739	0.747	0.746	0.743	0.721
fitness_club	IID	0.813	0.816	0.815	0.821	0.820	0.777	0.815
hazelnut_spread_contaminant_de	IID	0.943	0.939	0.940	0.994	0.995	0.958	0.953
heart_disease_cleveland	IID	0.894	0.892	0.897	0.908	0.908	0.902	0.892
heart_disease_hungary	IID	0.904	0.902	0.906	0.910	0.911	0.884	0.905
heart_disease_va_long_beach	IID	0.656	0.658	0.650	0.676	0.665	0.674	0.656
heart_failure_followup_surviva	IID	0.905	0.900	0.905	0.909	0.916	0.911	0.864
heloc	IID	0.792	0.790	0.792	0.802	0.802	0.792	0.781
hepatitis_survival_prediction	IID	0.867	0.866	0.867	0.857	0.868	0.880	0.817
homeq_default_prediction	IID	0.924	0.919	0.919	0.999	0.996	0.961	0.902
indian_liver_patient_dataset	IID	0.733	0.734	0.737	0.758	0.764	0.744	0.745
iranian_churn	IID	0.972	0.971	0.970	0.997	0.996	0.983	0.932
jm1	IID	0.716	0.717	0.717	0.773	0.770	0.749	0.713
ljubljana_breast_cancer	IID	0.702	0.707	0.713	0.710	0.712	0.661	0.685

Continued from previous page

Task	Split	O-L	F-L	C-L	PFN3	ICL2	RF	LR
marketing_campaign	IID	0.893	0.884	0.887	0.922	0.926	0.888	0.883
seismic_bumps	IID	0.772	0.777	0.775	0.782	0.779	0.747	0.754
south_africa_coronary_heart_di	IID	0.750	0.751	0.752	0.765	0.770	0.727	0.771
thyroid_discordant	IID	0.906	0.911	0.907	0.987	0.989	0.976	0.822
tour_travels_churn	IID	0.931	0.930	0.928	0.963	0.960	0.947	0.835
parkinsons_biomedical_voice_me	G	0.772	0.755	0.754	0.708	0.737	0.802	0.754
biomechanical_orthopaedic_pred	IID	0.953	0.954	0.954	0.960	0.960	0.950	0.957
ecoli_proteins	IID	0.954	0.953	0.953	0.973	0.974	0.971	0.969
hepatitis_c_prediction	IID	0.977	0.980	0.978	0.985	0.985	0.973	0.951
horse_colic_survival	IID	0.797	0.802	0.807	0.798	0.806	0.794	0.743
maternal_health_risk	IID	0.881	0.881	0.880	0.957	0.958	0.945	0.795
website_phishing	IID	0.951	0.949	0.951	0.982	0.982	0.972	0.859
cardiotocography	G	0.960	0.960	0.958	0.974	0.976	0.970	0.946
dementia_prediction	G	0.843	0.841	0.845	0.829	0.835	0.819	0.867
ghanas_indigenous_intel	T	0.853	0.825	0.828	0.905	0.888	0.718	0.780

Table 11 gives the equal-task summary aggregates, and Table 14 lists the task manifest. Tables 12–13 give model scores by target and split; Tables 15–20 give the divided per-task excess CE, accuracy gain, and AUC tables, retaining each available native size and selected conventional/published comparators. The sparse grouped and temporal slices cannot support general shift claims. Figure 10 gives the native-only rank comparison.

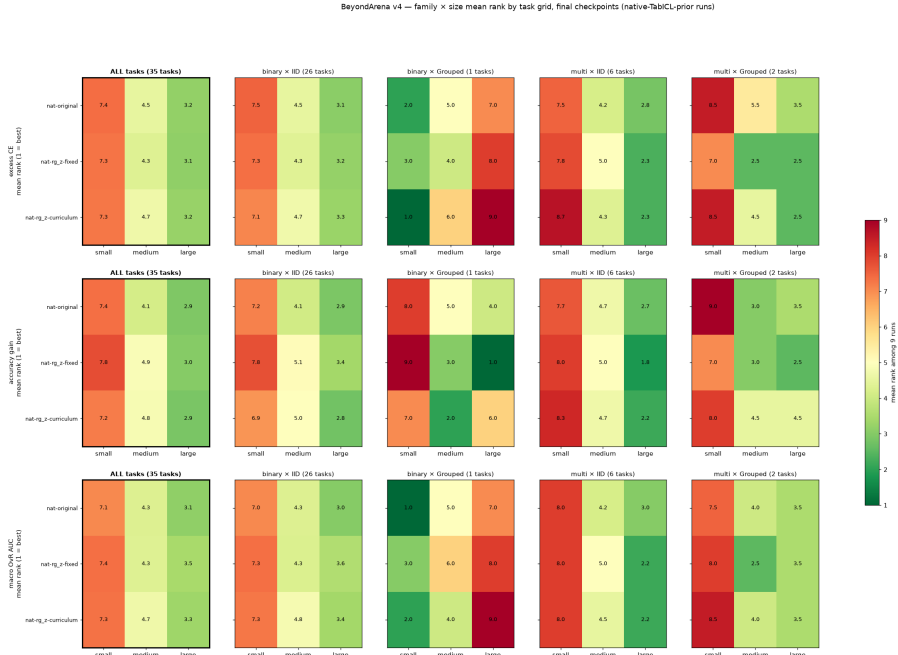


Figure 10: Mean task ranks among nine native-prior NanoTabPFN runs only. The overall rank uses 35 tasks and excludes the temporal task; published and conventional models are not in this ranking. Score tables in this appendix use all 36 tasks.

G MERGED HEART-SITE CASE STUDY

We merge the Cleveland (303 rows), Hungary (294), and VA Long Beach (200) records from the UCI Heart Disease source after confirming a shared 13-attribute schema and binary target (Janosi et al., 1989). The resulting 797-row dataset has an observed hospital label. Repeated stratified five-fold IID evaluation ($\times 2$) mixes sites in support and query, with 638 support rows per fold. Models receive either the 13 attributes alone (*site hidden*) or those attributes plus site (*site given*). A random forest predicts site from the 13 attributes at 0.918 five-fold accuracy (majority baseline 0.380), or 0.888 after median imputation removes much of the missingness cue; this makes site largely redundant in IID folds (Table 21).

For a target-site adaptation probe, all rows from the other two sites and 20 stratified labelled rows from the target site form support; the remaining target-site rows are queries. We draw five matched support subsets per target site, producing 15 folds with all three site categories represented in support. Frozen models receive (i) 13 attributes, (ii) attributes plus correct site tags, or (iii) attributes plus randomly permuted *support* site tags, with true query tags. The last two conditions have the same input width and support tag counts. Their contrast tests the usefulness of tag alignment, although a single random permutation per fold and overlapping repeated queries limit precision. All models use matched folds and the BeyondArena preprocessing path; Table 22 reports the results.

Table 21: Merged heart-site excess CE under repeated stratified IID folds (five folds, two repeats). All sites occur in support. Hidden uses 13 clinical attributes; given adds the observed site. Lower values are better. TabICL v1.1 was not evaluated in this case study.

Model	Site hidden	Site given
Native single-regime, 2 layers	-0.2308	-0.2298
Native fixed, 2 layers	-0.2251	-0.2229
Native curriculum, 2 layers	-0.2286	-0.2281
Native single-regime, 4 layers	-0.2737	-0.2753
Native fixed, 4 layers	-0.2722	-0.2749
Native curriculum, 4 layers	-0.2714	-0.2740
Native single-regime, 6 layers	-0.2657	-0.2665
Native fixed, 6 layers	-0.2659	-0.2663
Native curriculum, 6 layers	-0.2654	-0.2656
Random forest	-0.2637	-0.2639
Logistic regression	-0.2730	-0.2740
Histogram gradient boosting	-0.1480	-0.1436
CatBoost	-0.1731	-0.1689
XGBoost	-0.0609	-0.0589
LightGBM	+0.2753	+0.2795
TabICL v1	-0.2741	-0.2733
TabICL v2	-0.2836	-0.2833
TabPFN v2.2	-0.2820	-0.2822
TabPFN v2.6	-0.2839	-0.2845
TabPFN v3	-0.2841	-0.2825

Under IID folds, site changes CE by at most 0.0045 nats across the 20 evaluated models and has no consistent direction. In target-site adaptation, the aligned-minus-shuffled contrast in excess CE ranges from -0.0127 to $+0.0042$ nats, with 16 of 20 models negative; the largest aligned-tag gains are for TabICL v1, random forest, and the medium native configurations, and it does not reveal a consistent extra advantage from multiregime pretraining across sizes. The site is a known observed group, but this comparison does not establish a change of label mechanism at each site, nor latent-regime recovery.

Table 22: Target-site adaptation excess CE over 15 matched folds: five stratified 20-shot draws for each of three target sites. Support also contains all rows from the other two sites; queries are the remaining target-site rows. Shuffled permutes support-site tags while keeping tag counts and query tags fixed. True and shuffled have equal input width. Lower CE is better; negative true-minus-shuffled values favor aligned site tags.

Model	Hidden	Shuffled	True	True—shuffled
Native single-regime, 2 layers	-0.1664	-0.1692	-0.1651	+0.0041
Native fixed, 2 layers	-0.1746	-0.1794	-0.1769	+0.0025
Native curriculum, 2 layers	-0.1606	-0.1651	-0.1610	+0.0042
Native single-regime, 4 layers	-0.2519	-0.2503	-0.2608	-0.0105
Native fixed, 4 layers	-0.2520	-0.2567	-0.2683	-0.0117
Native curriculum, 4 layers	-0.2426	-0.2457	-0.2564	-0.0107
Native single-regime, 6 layers	-0.2661	-0.2670	-0.2714	-0.0044
Native fixed, 6 layers	-0.2622	-0.2640	-0.2667	-0.0027
Native curriculum, 6 layers	-0.2502	-0.2515	-0.2513	+0.0002
Random forest	-0.2472	-0.2439	-0.2556	-0.0117
Logistic regression	-0.2332	-0.2306	-0.2371	-0.0064
Histogram gradient boosting	-0.1731	-0.1704	-0.1775	-0.0071
CatBoost	-0.1540	-0.1645	-0.1668	-0.0023
XGBoost	-0.0817	-0.0809	-0.0892	-0.0083
LightGBM	+0.1996	+0.1874	+0.1808	-0.0066
TabICL v1	-0.2799	-0.2696	-0.2823	-0.0127
TabICL v2	-0.2857	-0.2843	-0.2949	-0.0106
TabPFN v2.2	-0.2891	-0.2905	-0.2931	-0.0026
TabPFN v2.6	-0.2940	-0.2954	-0.3038	-0.0084
TabPFN v3	-0.2879	-0.2865	-0.2935	-0.0070